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Fed-GEE: Federated Generalized Estimating Equations for Multisite Health Data, with an Application to Veterans’ Health

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Supporting Information for “Fed-GEE: Federated Generalized Estimating Equations for Multisite Health Data, with an Application to Veterans’ Health”

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Sections are the Web Appendices cited in the main text. Web Tables are numbered S1, S2, and so on.

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A Large-sample theory

Let

$$\mathcal{S}_K(\beta) = \sum_{i=1}^K \mathcal{S}_{iK}(\beta), \quad \dot{\mathcal{S}}_K(\beta) = \frac{\partial \mathcal{S}_K(\beta)}{\partial \beta^\top}.$$

Theorem A.1 (Consistency and asymptotic normality). *Suppose (i) sites are independent and $E\{\mathcal{S}_{iK}(\beta_0)\} = 0$; (ii) for a deterministic continuous function s ,*

$$\sup_{\beta \in \Theta} \|K^{-1} \mathcal{S}_K(\beta) - s(\beta)\| \xrightarrow{p} 0,$$

and β_0 is a separated zero of s ; (iii), on a neighborhood of β_0 ,

$$\sup_{\beta} \|-K^{-1} \dot{\mathcal{S}}_K(\beta) - B(\beta)\| \xrightarrow{p} 0,$$

where B is continuous and $B_0 = B(\beta_0)$ is nonsingular; and (iv)

$$K^{-1/2} \mathcal{S}_K(\beta_0) \xrightarrow{d} N(0, M_0).$$

If $\hat{\beta}$ is the selected root of $\mathcal{S}_K$, then $\hat{\beta} \xrightarrow{p} \beta_0$ and

$$\sqrt{K}(\hat{\beta} - \beta_0) \xrightarrow{d} N\{0, B_0^{-1} M_0 B_0^{-\top}\}.$$

Proof. Uniform convergence and separated identification imply consistency. The integral Taylor identity gives

$$0 = \mathcal{S}_K(\beta_0) + \left\{ \int_0^1 \dot{\mathcal{S}}_K(\beta_0 + t(\hat{\beta} - \beta_0)) dt \right\} (\hat{\beta} - \beta_0).$$

The negative normalized integral Jacobian converges in probability to B_0 . Premultiplication by its inverse followed by Slutsky's theorem gives the stated limit. $\square$

B Exactness of the half-power

All expectations in this section are conditional on the design.

Theorem B.1 (Exact score-space correction). *Suppose*

$$\mathcal{S}_i(\beta) = U_i - B_i(\beta - \beta_0),$$

where B_i is fixed and positive semidefinite, the U_i are independent, $E(U_i) = 0$, and $\text{Var}(U_i) = \tau B_i$ for a common $\tau > 0$. Let $B = \sum_i B_i \succ 0$ and assume $B - B_i \succ 0$ for every i . Define

$$G_i = B^{-1/2} B_i B^{-1/2}, \quad z_i = B^{-1/2} \mathcal{S}_i(\hat{\beta}), \quad A_i = (\mathbf{I} - G_i)^{-1/2}.$$

Then

$$E(z_i z_i^\top) = \tau G_i (\mathbf{I} - G_i)$$

and

$$\hat{V}_{\text{KC}} = B^{-1/2} \left\{ \sum_i A_i z_i z_i^\top A_i \right\} B^{-1/2}$$

satisfies

$$E(\hat{V}_{\text{KC}}) = \tau B^{-1} = \text{Var}(\hat{\beta}).$$

Proof. The estimating equation gives

$$\hat{\beta} - \beta_0 = B^{-1} \sum_j U_j.$$

Writing $H_i = B_i B^{-1}$,

$$\mathcal{S}_i(\hat{\beta}) = (\mathbf{I} - H_i) U_i - H_i \sum_{j \neq i} U_j.$$

Independence and the covariance condition therefore imply

$$\begin{aligned} E\{\mathcal{S}_i(\hat{\beta}) \mathcal{S}_i(\hat{\beta})^\top\} &= (\mathbf{I} - H_i)(\tau B_i)(\mathbf{I} - H_i)^\top \\ &\quad + H_i\{\tau(B - B_i)\}H_i^\top. \end{aligned}$$

Conjugation by $B^{-1/2}$ yields

$$E(z_i z_i^\top) = \tau G_i (\mathbf{I} - G_i).$$

Because A_i is a matrix function of G_i , the two commute, and

$$A_i E(z_i z_i^\top) A_i = \tau G_i.$$

Finally, $\sum_i G_i = \mathbf{I}$, so

$$E(\hat{V}_{\text{KC}}) = B^{-1/2} \left(\tau \sum_i G_i \right) B^{-1/2} = \tau B^{-1}.$$

The same value follows directly for the variance of $B^{-1} \sum_i U_i$. □

B.1 Uniqueness of the half-power

For $A_i(\alpha) = (\mathbf{I} - G_i)^{-\alpha}$, the expected adjusted information-standardized contribution is

$$\tau G_i (\mathbf{I} - G_i)^{1-2\alpha}.$$

Requiring this to equal τG_i for every admissible G_i forces $1 - 2\alpha = 0$. Thus $\alpha = 1/2$ is unique within this family. The full-power choice $\alpha = 1$ is larger in positive-semidefinite order under the affine working model and is consequently expected to be conservative.

B.2 Raw-score representation

The equivalent adjusted raw score is

$$\tilde{\mathcal{S}}_i = B^{1/2} (\mathbf{I} - G_i)^{-1/2} B^{-1/2} \mathcal{S}_i(\hat{\beta}),$$

and

$$\hat{V}_{\text{KC}} = B^{-1} \left(\sum_i \tilde{\mathcal{S}}_i \tilde{\mathcal{S}}_i^\top \right) B^{-1}.$$

This representation shows why the correction cannot generally be implemented by multiplying a raw score directly by $(\mathbf{I} - G_i)^{-1/2}$.

C Invariance to the matrix square root

All three corrections in the main text use the symmetric square root $\hat{\mathcal{B}}^{-1/2}$. Any other factor gives the same variance.

Proposition C.1 (Invariance to the choice of matrix factor). *Let $B \succ 0$ and let R be any $p \times p$ matrix with $RR^\top = B^{-1}$. Define*

$$G_i^R = R^\top B_i R, \quad z_i^R = R^\top \mathcal{S}_i(\hat{\beta}), \quad A_i^R(\alpha) = (\mathbf{I} - G_i^R)^{-\alpha}.$$

Then, for every $\alpha > 0$,

$$R \left\{ \sum_{i=1}^K A_i^R(\alpha) z_i^R z_i^{R\top} A_i^R(\alpha)^\top \right\} R^\top = B^{-1/2} \left\{ \sum_{i=1}^K A_i(\alpha) z_i z_i^\top A_i(\alpha) \right\} B^{-1/2}.$$

Proof. Let $Q = B^{1/2} R$. Then $QQ^\top = B^{1/2} B^{-1} B^{1/2} = \mathbf{I}$, so Q is orthogonal and $R = B^{-1/2} Q$. Hence $G_i^R = Q^\top G_i Q$ and $z_i^R = Q^\top z_i$. Because Q is orthogonal, $(\mathbf{I} - Q^\top G_i Q)^{-\alpha} =$

$Q^\top(\mathbf{I} - G_i)^{-\alpha}Q$. Substituting, each term equals $B^{-1/2}QQ^\top A_iQQ^\top z_i z_i^\top QQ^\top A_iQQ^\top B^{-1/2} = B^{-1/2}A_i z_i z_i^\top A_i B^{-1/2}$. $\square$

A Cholesky factor of B^{-1} may therefore be used in computation. The same argument shows that the degrees of freedom of Web Appendix F do not depend on the factor, because they depend on the design only through the corrected raw-score operators T_i , which are factor invariant.

D Large- K equivalence of the corrections

Proposition D.1 (Large- K equivalence of the corrections). *Let $B_K = \hat{\mathcal{B}}$, $q_K = \max_i \|G_i\|$, and $z_i = B_K^{-1/2}\hat{\mathcal{S}}_i$. Suppose $q_K = O_p(K^{-1})$, $\sum_i \|z_i\|^2 = O_p(1)$, and $\|B_K^{-1/2}\| = O_p(K^{-1/2})$. Then $\|\hat{V}_{\text{KC}} - \hat{V}_{\text{S}}\| = O_p(K^{-2})$, while $\|\hat{V}_{\text{S}}\| = O_p(K^{-1})$.*

The condition $q_K = O_p(K^{-1})$ says that no site holds more than a vanishing share of the information.

Proof. Let

$$A_i(1/2) = (\mathbf{I}_p - G_i)^{-1/2}, \quad q_K = \max_{1 \leq i \leq K} \|G_i\|.$$

The scalar expansion

$$(1 - x)^{-1/2} = \sum_{m=0}^{\infty} c_m x^m, \quad c_m = \frac{\binom{2m}{m}}{4^m},$$

induces a convergent matrix power series whenever $\|G_i\| < 1$. Because $q_K = O_p(K^{-1})$, with probability tending to one $q_K \leq 1/2$. On this event,

$$d_K := \max_i \|A_i(1/2) - \mathbf{I}_p\| \leq \sum_{m=1}^{\infty} c_m q_K^m = (1 - q_K)^{-1/2} - 1 \leq C q_K$$

for a finite constant C . Hence

$$d_K = O_p(K^{-1}), \quad \max_i \|A_i(1/2)\| = O_p(1).$$

Using the identity

$$\begin{aligned} & A_i(1/2) z_i z_i^\top A_i(1/2) - z_i z_i^\top \\ &= \{A_i(1/2) - \mathbf{I}_p\} z_i z_i^\top A_i(1/2) + z_i z_i^\top \{A_i(1/2) - \mathbf{I}_p\}, \end{aligned}$$

submultiplicativity and $\|z_i z_i^\top\| = \|z_i\|^2$ give

$$\begin{aligned} & \left\| \sum_{i=1}^K \left(A_i(1/2) z_i z_i^\top A_i(1/2) - z_i z_i^\top \right) \right\| \\ & \leq d_K \left\{ \max_i \|A_i(1/2)\| + 1 \right\} \sum_{i=1}^K \|z_i\|^2 = O_p(K^{-1}). \end{aligned}$$

Therefore, using $\|B_K^{-1/2}\| = O_p(K^{-1/2})$,

$$\|\hat{V}_{\text{KC}} - \hat{V}_{\text{S}}\| \leq \|B_K^{-1/2}\|^2 O_p(K^{-1}) = O_p(K^{-2}).$$

Moreover,

$$\|\hat{V}_{\text{S}}\| \leq \|B_K^{-1/2}\|^2 \sum_{i=1}^K \|z_i\|^2 = O_p(K^{-1}).$$

Thus the half-power correction is second order relative to the K^{-1} scale of the conventional sandwich covariance and does not alter its first-order large- K limit. $\square$

Remark D.2. The same argument applies to MD, with the coefficients c_m replaced by 1, and to FG, whose diagonal factors satisfy $\|F_i - \mathbf{I}\| = O_p(K^{-1})$ when the condition number of $K^{-1}B_K$ is bounded. All corrections therefore share the first-order limit of the uncorrected sandwich.

E Rank of the fitted meat

For the unadjusted fitted score matrix $S = [\mathcal{S}_1(\hat{\beta}), \dots, \mathcal{S}_K(\hat{\beta})]$, the estimating equation gives $S\mathbf{1}_K = 0$. Consequently,

$$\text{rank}(SS^\top) \leq \min(p, K - 1).$$

Site-specific invertible adjustments need not preserve the zero-sum relation. The universal adjusted bound is instead

$$\text{rank} \left(\sum_i A_i \mathcal{S}_i \mathcal{S}_i^\top A_i^\top \right) \leq \min(p, K).$$

Thus an adjusted meat is necessarily singular when $K < p$, but it can be full rank when $K = p$. This distinction does not rescue routine Wald inference at $K \leq p$, where the uncorrected score covariance and residual degrees of freedom are inadequate.

F Fay–Graubard and Bell–McCaffrey in score space

F.1 Fay–Graubard in score space

Fay and Graubard [1] rescale each cluster’s contribution to the meat by a diagonal factor built from the cluster’s share of the information. At the site level this factor is

$$F_i = \text{diag} \left[\left\{ 1 - \min \left(b, (B_i B^{-1})_{kk} \right) \right\}^{-1/2} \right]_k^p,$$

with $b = 0.75$ by default. The corrected variance is $B^{-1} \left(\sum_i F_i \hat{\mathcal{S}}_i \hat{\mathcal{S}}_i^\top F_i \right) B^{-1}$. It uses only the site scores and information matrices. Our implementation agreed with the **saws** package (method **d5**) to machine precision when **saws** was given the same site scores and information matrices.

F.2 Bell–McCaffrey degrees of freedom in score space

Fix a coefficient j and a correction with raw-score operators T_i (main text, equation (9)). Let $a_i = T_i^\top B^{-1} e_j$. The corrected variance of $\hat{\beta}_j$ is

$$v_j = \sum_{i=1}^K (a_i^\top \hat{\mathcal{S}}_i)^2.$$

Work under the affine model of Theorem 1 of the main text, with $U_k = \tau^{1/2} B_k^{1/2} \varepsilon_k$ and independent standard normal ε_k . Then $\hat{\mathcal{S}}_i = U_i - B_i B^{-1} \sum_k U_k$, so

$$a_i^\top \hat{\mathcal{S}}_i = \sum_{k=1}^K w_{ik}^\top U_k, \quad w_{ik} = \mathbf{1}\{i = k\} a_i - B^{-1} B_i a_i.$$

Hence $v_j = \tau \varepsilon^\top Q \varepsilon$ for a positive semidefinite matrix $Q = \sum_i r_i r_i^\top$, where r_i stacks the blocks $B_k^{1/2} w_{ik}$. The nonzero eigenvalues of Q equal those of the $K \times K$ matrix C with entries $C_{il} = r_i^\top r_l = \sum_k w_{ik}^\top B_k w_{lk}$. Expanding and using $\sum_k B_k = B$,

$$C_{il} = \mathbf{1}\{i = l\} a_i^\top B_i a_i - (B_i a_i)^\top B^{-1} (B_l a_l).$$

For a Gaussian quadratic form, $E(v_j) = \tau \text{tr}(C)$ and $\text{Var}(v_j) = 2\tau^2 \text{tr}(C^2)$. Matching these two moments to a scaled χ_ν^2 distribution [2, 3] gives

$$\nu_j^{\text{BM}} = \frac{\{\text{tr}(C)\}^2}{\text{tr}(C^2)}.$$

Equal sites. If $B_i = B/K$ and $T_i = T$ for all i , then $a_i = a$ and $C = c(\mathbf{I}_K - K^{-1}\mathbf{1}_K\mathbf{1}_K^\top)$ with $c = a^\top Ba/K$. Then $\text{tr}(C) = c(K-1)$ and $\text{tr}(C^2) = c^2(K-1)$, so $\nu_j^{\text{BM}} = K-1$.

Relation to CR2. With working independence and an identity link, the site information matrices are $X_i^\top X_i$, the KC operator equals the CR2 adjustment, and ν_j^{BM} equals the Satterthwaite degrees of freedom of Pustejovsky and Tipton [4]. We confirmed agreement to four decimal places against the `clubSandwich` package in simulated data. In the VHA application, with a logistic link, the degrees of freedom agreed to three decimal places and the standard errors to within 0.5%.

Monte Carlo check. Under the working model with 20,000 simulated draws, the empirical mean and variance of v_j matched $\tau \text{tr}(C)$ and $2\tau^2 \text{tr}(C^2)$.

G One-step Fed-GEE

When communication rounds are costly, a common initial estimator $\hat{\beta}^{(0)}$ can be updated once using

$$\hat{\beta}^{(1)} = \hat{\beta}^{(0)} + \mathcal{B}_K\{\hat{\beta}^{(0)}\}^{-1} \mathcal{S}_K\{\hat{\beta}^{(0)}\}. \quad (\text{G.1})$$

Each site sends one score and one information matrix evaluated at the common initializer.

Assumption G.1 (Initial rate). *The common initializer satisfies*

$$\hat{\beta}^{(0)} - \beta_0 = o_p(K^{-1/4}).$$

Assumption G.2 (Local smoothness). *In a neighborhood of β_0 ,*

$$\|B_i(\beta) - B_i(\beta')\| \leq \ell_i \|\beta - \beta'\|, \quad K^{-1} \sum_i \ell_i = O_p(1),$$

and the normalized difference between $B_K(\beta)$ and the negative score derivative is $O_p(K^{-1/2})$ uniformly in that neighborhood.

Theorem G.1 (One-step Fed-GEE). *Under the centralized conditions and the two preceding assumptions,*

$$\sqrt{K}(\hat{\beta}^{(1)} - \beta_0) \xrightarrow{d} N\{0, B_0^{-1} M_0 B_0^{-\top}\}.$$

Thus the one-step and fully converged estimators have the same first-order limit.

Proof. Expand $\mathcal{S}_K\{\hat{\beta}^{(0)}\}$ around β_0 using an integral Jacobian. After substituting into Equation (G.1), the linear term cancels the initial error. The bread Lipschitz condition

bounds the remaining curvature term by

$$O_p(1)\sqrt{K}\left\|\hat{\beta}^{(0)} - \beta_0\right\|^2 = o_p(1).$$

The normalized bread–score-derivative discrepancy contributes $O_p(1)\left\|\hat{\beta}^{(0)} - \beta_0\right\| = o_p(1)$. Hence

$$\{K^{-1}\mathcal{B}_K(\hat{\beta}^{(0)})\}\sqrt{K}(\hat{\beta}^{(1)} - \beta_0) = K^{-1/2}\mathcal{S}_K(\beta_0) + o_p(1).$$

The bread converges to B_0 , and the site-level central limit theorem and Slutsky’s theorem give the result. $\square$

The rate condition is substantive. A one-shot update is not recommended when a common $o_p(K^{-1/4})$ initializer cannot be justified.

H Decentralized Fed-GEE

H.1 Communication Conditions

In the decentralized architecture, each site exchanges low-dimensional quantities only with its network neighbors and approximates global summaries through average consensus. The centralized regularity conditions stated in the main manuscript continue to govern the statistical model; the conditions below concern communication and numerical approximation.

Condition H.1 (Communication network). *For each K , let*

$$W_K = (w_{ih}) \in \mathbb{R}^{K \times K}$$

denote the mixing matrix, where i and h index sites. The entries of W_K are nonnegative, and W_K is symmetric and doubly stochastic:

$$W_K \mathbf{1}_K = \mathbf{1}_K, \quad \mathbf{1}_K^\top W_K = \mathbf{1}_K^\top.$$

The corresponding undirected communication graph is connected, with

$$w_{ih} = 0$$

whenever sites i and h are not neighbors and $i \neq h$. Define

$$J_K = \frac{1}{K} \mathbf{1}_K \mathbf{1}_K^\top.$$

Assume the mixing process is aperiodic, for example through positive diagonal weights, so that

$$\rho_K = \|W_K - J_K\|_2 < 1.$$

For a site-level quantity Q , define

$$\delta_{Q,K} = \|W_K^{L_{Q,K}} - J_K\|_\infty, \quad (\text{H.1})$$

where $L_{Q,K}$ is the number of consensus rounds applied to Q .

Condition H.2 (Site-level smoothness and boundedness). For all $\beta_1, \beta_2 \in \Theta$,

$$\|S_h(\beta_1) - S_h(\beta_2)\|_2 \leq \ell_{S,h,K} \|\beta_1 - \beta_2\|_2,$$

where

$$\frac{1}{K} \sum_{h=1}^K \ell_{S,h,K} = O_p(1).$$

Similarly,

$$\|B_h(\beta_1) - B_h(\beta_2)\|_2 \leq \ell_{B,h,K} \|\beta_1 - \beta_2\|_2,$$

where

$$\frac{1}{K} \sum_{h=1}^K \ell_{B,h,K} = O_p(1).$$

In addition,

$$\max_h \sup_{\beta \in \Theta} \|S_h(\beta)\|_2 = O_p(1), \quad \max_h \sup_{\beta \in \Theta} \|B_h(\beta)\|_2 = O_p(1).$$

Let $\beta_h^\star$ denote the terminal local parameter state at site h immediately before the final parameter-consensus step. Assume

$$\max_h \|\beta_h^\star\|_2 = O_p(1).$$

Condition H.3 (Terminal algorithmic accuracy). After the final parameter-consensus step, site i holds

$$\hat{\beta}_i = \sum_{h=1}^K [W_K^{L_{\beta,K}}]_{ih} \beta_h^\star. \quad (\text{H.2})$$

At $\hat{\beta}_h$, each site recomputes its local score and bread. After score and bread consensus, site i holds

$$\hat{S}_i = \sum_{h=1}^K [W_K^{L_{S,K}}]_{ih} S_h(\hat{\beta}_h),$$

and

$$\widehat{B}_i = \sum_{h=1}^K [W_K^{L_{B,K}}]_{ih} B_h(\widehat{\beta}_h).$$

Assume $\widehat{B}_i$ is nonsingular with probability tending to one. Define the achieved Fisher-scoring residual by

$$\eta_K = \max_i \left\| \widehat{B}_i^{-1} \widehat{S}_i \right\|_2. \quad (\text{H.3})$$

The algorithm is terminated only when

$$\eta_K \leq \epsilon_{\eta,K}.$$

For consistency, require $\epsilon_{\eta,K} \rightarrow 0$; for root- K asymptotic equivalence, require

$$\epsilon_{\eta,K} = o(K^{-1/2}).$$

H.2 Estimator and Consensus Approximation

Define the decentralized estimator by the network average

$$\widehat{\beta}_{\text{dec}} = \frac{1}{K} \sum_{i=1}^K \widehat{\beta}_i. \quad (\text{H.4})$$

Because W_K is doubly stochastic,

$$\widehat{\beta}_{\text{dec}} = \frac{1}{K} \sum_{h=1}^K \beta_h^*.$$

Thus the final parameter-consensus step preserves the network average.

Lemma H.1 (Average-consensus approximation). *Let $Q_1, \dots, Q_K$ be site-specific vectors or matrices satisfying*

$$\max_h \|Q_h\| = O_p(1).$$

After $L_{Q,K}$ consensus rounds, let

$$\widehat{Q}_i = \sum_{h=1}^K [W_K^{L_{Q,K}}]_{ih} Q_h.$$

Then

$$\max_i \left\| \widehat{Q}_i - \frac{1}{K} \sum_{h=1}^K Q_h \right\| = O_p(\delta_{Q,K}).$$

Lemma H.2 (Final parameter agreement). *Under Conditions H.1 and H.2,*

$$\max_i \left\| \hat{\beta}_i - \hat{\beta}_{\text{dec}} \right\|_2 = O_p(\delta_{\beta,K}).$$

Lemma H.3 (Approximate global estimating equation). *Under Conditions H.1–H.3,*

$$\left\| \bar{S}_K(\hat{\beta}_{\text{dec}}) \right\|_2 = O_p(r_K),$$

where

$$r_K = \delta_{\beta,K} + \delta_{S,K} + \epsilon_{\eta,K}. \quad (\text{H.5})$$

H.3 Large-Sample Properties

Theorem H.4 (Consistency and asymptotic equivalence). *Suppose the centralized consistency conditions stated in the main manuscript hold, together with Conditions H.1–H.3. If*

$$r_K = o(1),$$

then

$$\hat{\beta}_{\text{dec}} \xrightarrow{p} \beta_0.$$

Moreover,

$$\left\| \hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}} \right\|_2 = O_p(r_K).$$

Theorem H.5 (Asymptotic normality). *Suppose the centralized asymptotic-normality conditions stated in the main manuscript hold, together with Conditions H.1–H.3. If*

$$r_K = o(K^{-1/2}),$$

then

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \beta_0) \xrightarrow{d} N(\mathbf{0}, B_0^{-1} M_0 B_0^{-\top}).$$

Corollary H.6 (Sufficient communication and stopping rates). *Consistency follows if*

$$\delta_{\beta,K} \rightarrow 0, \quad \delta_{S,K} \rightarrow 0, \quad \epsilon_{\eta,K} \rightarrow 0.$$

Root-K asymptotic equivalence follows if

$$\delta_{\beta,K} + \delta_{S,K} + \epsilon_{\eta,K} = o(K^{-1/2}).$$

If the final global bread is also reconstructed by consensus, require

$$\delta_{B,K} \rightarrow 0.$$

For the finite-sample score correction, the fitted leverage matrices must remain well defined, with

$$\lambda_{\max}(\hat{G}_i) < 1$$

in parameter directions in which a negative matrix power is used.

H.4 Consensus Rounds

For any site-level quantity Q_i , let $Q_i^{[0]} = Q_i$. At round ℓ ,

$$Q_i^{[\ell+1]} = \sum_{h=1}^K w_{ih} Q_h^{[\ell]}.$$

Stacking the site states gives

$$Q^{[\ell+1]} = W_K Q^{[\ell]},$$

and after L rounds,

$$Q^{[L]} = W_K^L Q^{[0]}.$$

For a symmetric mixing matrix,

$$\|W_K^L - J_K\|_2 = \rho_K^L, \quad \rho_K = \|W_K - J_K\|_2 < 1.$$

Because, for any $K \times K$ matrix C ,

$$\|C\|_\infty \leq \sqrt{K} \|C\|_2,$$

the row-sum consensus error satisfies

$$\|W_K^L - J_K\|_\infty \leq \sqrt{K} \rho_K^L. \tag{H.6}$$

Thus increasing the number of communication rounds reduces the worst-case consensus bound geometrically. The realized error at an individual site need not decrease monotonically after every single additional round because network eigen-components can temporarily cancel or reinforce one another.

H.5 Finite-Sample Score Correction in the Decentralized Architecture

After final parameter consensus, site i evaluates

$$S_i(\hat{\beta}_i), \quad B_i(\hat{\beta}_i).$$

After L rounds of bread consensus, let

$$\hat{B}_i^{(L)} = \sum_{h=1}^K [W_K^L]_{ih} B_h(\hat{\beta}_h),$$

and reconstruct the network-wide bread sum by

$$\hat{B}_{K,i}^{(L)} = K \hat{B}_i^{(L)}. \quad (\text{H.7})$$

Site i then forms

$$\hat{G}_i^{(L)} = \{\hat{B}_{K,i}^{(L)}\}^{-1/2} B_i(\hat{\beta}_i) \{\hat{B}_{K,i}^{(L)}\}^{-1/2}, \quad (\text{H.8})$$

and, for $a \in \{0, 1/2, 1\}$,

$$\hat{A}_i^{(L)}(a) = (\mathbf{I}_p - \hat{G}_i^{(L)})^{-a}. \quad (\text{H.9})$$

The corrected local score is

$$S_i^{\text{corr},(L)}(a) = \{\hat{B}_{K,i}^{(L)}\}^{1/2} \hat{A}_i^{(L)}(a) \{\hat{B}_{K,i}^{(L)}\}^{-1/2} S_i(\hat{\beta}_i), \quad (\text{H.10})$$

with local corrected meat contribution

$$M_i^{\text{corr},(L)}(a) = S_i^{\text{corr},(L)}(a) S_i^{\text{corr},(L)}(a)^\top.$$

These local meat contributions can be aggregated through an additional consensus step.

Let

$$\hat{B}_K^{\text{dec}} = \sum_{h=1}^K B_h(\hat{\beta}_h).$$

By Lemma H.1,

$$\left\| \hat{B}_i^{(L)} - \frac{1}{K} \hat{B}_K^{\text{dec}} \right\| = O_p(\delta_{B,K}^{(L)}),$$

so

$$\left\| \hat{B}_{K,i}^{(L)} - \hat{B}_K^{\text{dec}} \right\| = O_p\{K \delta_{B,K}^{(L)}\}.$$

Since the global bread is of order K , this corresponds to a relative error of order $O_p(\delta_{B,K}^{(L)})$.

Under continuity of the inverse square-root and matrix-power maps, together with eigenvalue gaps away from zero, vanishing parameter- and bread-consensus errors imply that $\hat{G}_i^{(L)}$, $\hat{A}_i^{(L)}(a)$, and the corrected local meat contributions converge to their centralized counterparts.

H.5.1 Reference distributions from additive summaries

The Bell–McCaffrey degrees of freedom also admit a consensus formulation. For a fixed coefficient j , write $B = \sum_i B_i$, $a_i = T_i^\top B^{-1} e_j$, $d_i = a_i^\top B_i a_i$, and $u_i = B_i a_i$. Define the additive summaries

$$D_1 = \sum_i d_i, \quad D_2 = \sum_i d_i^2, \quad H = \sum_i u_i u_i^\top, \quad R = \sum_i d_i u_i u_i^\top.$$

The kernel in the main text is $C = D - U^\top B^{-1} U$, where $D = \text{diag}(d_1, \dots, d_K)$ and U has columns u_i . Cyclic invariance of the trace gives

$$\text{tr}(C) = D_1 - \text{tr}(B^{-1} H), \tag{H.11}$$

$$\text{tr}(C^2) = D_2 - 2 \text{tr}(B^{-1} R) + \text{tr}(B^{-1} H B^{-1} H). \tag{H.12}$$

Thus, no exchange of the full $K \times K$ kernel is required. After information consensus, each site constructs its local d_i, u_i using its approximation to B , and applies consensus to $d_i, d_i^2, u_i u_i^\top, d_i u_i u_i^\top$. Multiplication by K converts the consensus averages into approximate totals.

For fixed K , continuity of these expressions implies convergence to the coordinated degrees of freedom as parameter and summary consensus errors vanish, provided the information matrix remains nonsingular, the correction operators remain well defined, and $\text{tr}(C^2) > 0$ at the limit. The resulting critical values converge whenever the limiting degrees of freedom are positive. This is a computational equivalence result and does not establish exact finite-sample coverage. The decentralized simulations in Web Appendix I do not evaluate this extension.

H.6 Proofs for the Decentralized Results

H.6.1 Proof of Lemma H.1

Proof. For every site i ,

$$\hat{\bar{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h = \sum_{h=1}^K \left\{ [W_K^{L_{Q,K}}]_{ih} - \frac{1}{K} \right\} Q_h.$$

By the triangle inequality,

$$\left\| \hat{\bar{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h \right\| \leq \|W_K^{L_{Q,K}} - J_K\|_\infty \max_h \|Q_h\|.$$

Taking the maximum over i gives

$$\max_i \left\| \hat{\bar{Q}}_i - \frac{1}{K} \sum_{h=1}^K Q_h \right\| = O_p(\delta_{Q,K}).$$

□

H.6.2 Proof of Lemma H.2

Proof. Because W_K is doubly stochastic,

$$\hat{\beta}_{\text{dec}} = \frac{1}{K} \sum_{h=1}^K \beta_h^*.$$

Applying Lemma H.1 with $Q_h = \beta_h^*$ gives

$$\max_i \left\| \hat{\beta}_i - \hat{\beta}_{\text{dec}} \right\|_2 = O_p(\delta_{\beta,K}).$$

□

H.6.3 Proof of Lemma H.3

Proof. Fix a site i and decompose

$$\bar{S}_K(\hat{\beta}_{\text{dec}}) = R_{S,1} + R_{S,2,i} + R_{S,3,i},$$

where

$$R_{S,1} = \frac{1}{K} \sum_{h=1}^K \{S_h(\hat{\beta}_{\text{dec}}) - S_h(\hat{\beta}_h)\},$$

$$R_{S,2,i} = \frac{1}{K} \sum_{h=1}^K S_h(\hat{\beta}_h) - \hat{S}_i, \quad R_{S,3,i} = \hat{S}_i.$$

By site-level score Lipschitz continuity and Lemma H.2,

$$\|R_{S,1}\|_2 \leq \left(\frac{1}{K} \sum_{h=1}^K \ell_{S,h,K} \right) \max_h \|\hat{\beta}_h - \hat{\beta}_{\text{dec}}\|_2 = O_p(\delta_{\beta,K}).$$

Lemma H.1 applied to $Q_h = S_h(\hat{\beta}_h)$ gives

$$\max_i \|R_{S,2,i}\|_2 = O_p(\delta_{S,K}).$$

For the final term,

$$\hat{S}_i = \hat{B}_i(\hat{B}_i^{-1} \hat{S}_i).$$

The bounded-bread condition and convexity of the consensus weights give

$$\max_i \|\hat{B}_i\|_2 = O_p(1),$$

while the stopping rule gives

$$\max_i \|\hat{B}_i^{-1} \hat{S}_i\|_2 \leq \epsilon_{\eta,K}.$$

Hence

$$\max_i \|R_{S,3,i}\|_2 = O_p(\epsilon_{\eta,K}).$$

Combining terms yields

$$\|\bar{S}_K(\hat{\beta}_{\text{dec}})\|_2 = O_p(\delta_{\beta,K} + \delta_{S,K} + \epsilon_{\eta,K}) = O_p(r_K).$$

□

H.6.4 Proof of Theorem H.4

Proof. Lemma H.3 gives

$$\|\bar{S}_K(\hat{\beta}_{\text{dec}})\|_2 = O_p(r_K) = o_p(1).$$

Let $s(\beta)$ denote the deterministic uniform limit of $\bar{S}_K(\beta)$. Then

$$\|s(\hat{\beta}_{\text{dec}})\|_2 \leq \sup_{\beta \in \Theta} \|s(\beta) - \bar{S}_K(\beta)\|_2 + \|\bar{S}_K(\hat{\beta}_{\text{dec}})\|_2 = o_p(1).$$

Identification therefore implies

$$\hat{\beta}_{\text{dec}} \xrightarrow{p} \beta_0.$$

Since $\bar{S}_K(\hat{\beta}_{\text{cen}}) = \mathbf{0}$, a first-order expansion between the centralized and decentralized estimators gives

$$\bar{S}_K(\hat{\beta}_{\text{dec}}) = -\bar{B}_K^*(\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}) + o_p\left(\|\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}\|\right),$$

where $\bar{B}_K^* \xrightarrow{p} B_0$. Therefore,

$$\|\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}\|_2 = O_p(r_K).$$

□

H.6.5 Proof of Theorem H.5

Proof. From Theorem H.4,

$$\|\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}\|_2 = O_p(r_K).$$

If $r_K = o(K^{-1/2})$, then

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \hat{\beta}_{\text{cen}}) = o_p(1).$$

Therefore,

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \beta_0) = \sqrt{K}(\hat{\beta}_{\text{cen}} - \beta_0) + o_p(1).$$

The centralized asymptotic-normality result and Slutsky's theorem give

$$\sqrt{K}(\hat{\beta}_{\text{dec}} - \beta_0) \xrightarrow{d} N\left(\mathbf{0}, B_0^{-1} M_0 B_0^{-\top}\right).$$

□

I Decentralized simulation

This appendix reports a simulation of the decentralized algorithm of Web Appendix H. It was run before the correction and reference distribution of the main text were finalized. All intervals use the uncorrected site sandwich with a t_{K-p} reference. The simulation therefore shows how consensus error affects estimation and inference. It does not evaluate the KC correction or the Bell–McCaffrey degrees of freedom under consensus.

Design. Outcomes followed the latent Gaussian design of the main text, with patient-level correlation 0.30 and site-level correlation $\rho_H \in \{0, 0.1, 0.2\}$. Sites had 10 to 30 patients with 2 to 6 visits each. The number of sites was $K \in \{5, 10, 15, 25, 50, 100\}$, the average prevalence was 0.07 or 0.30, and some scenarios added a binary site-level covariate ($p = 3$). Each scenario had 200 to 400 replicates. Four networks were compared: a complete graph; a

regional network, in which sites form groups of five around a regional hub and the hubs are fully connected; a ring in which each site is linked to its two nearest neighbors on each side; and a single hub linked to every site. Mixing weights were Metropolis–Hastings weights. Each consensus stage used $L = 50$ rounds.

Results. Web Table S1 gives the spectral quantity ρ that governs consensus error, and Web Table S2 gives coverage and SE/SD by scenario. On the complete graph the decentralized results equalled the centralized results in every scenario. On the regional network they differed by at most 0.01 in coverage. The hub network matched the centralized results for $K \leq 25$, where $\rho^{50} \leq 0.13$. At $K = 50$ its coverage fell to 0.795 to 0.913, and at $K = 100$, where $\rho^{50} = 0.61$, to 0.060 to 0.580. The estimate stayed within 0.1 of the true slope of 1, but this error was up to 2.9 standard deviations, and the standard errors were too small (SE/SD 0.51 to 0.63). The ring had $\rho^{50} = 0.94$ at $K = 100$ and covered at 0.905 to 0.953, compared with 0.925 to 0.965 centrally. A fixed number of rounds is therefore adequate for some networks and not for others. The required number depends on ρ and grows as the network becomes less connected.

Table S1: Second-largest eigenvalue modulus ρ of the mixing matrix, and ρ^{50} , for each network and number of sites K .

K	Complete		Regional		Ring		Hub	
	ρ	ρ^{50}	ρ	ρ^{50}	ρ	ρ^{50}	ρ	ρ^{50}
5	0.000	0.000	0.000	0.000	0.539	0.000	0.800	0.000
10	0.000	0.000	0.950	0.078	0.873	0.001	0.900	0.005
15	0.000	0.000	0.944	0.055	0.942	0.051	0.933	0.032
25	0.000	0.000	0.941	0.049	0.979	0.347	0.960	0.130
50	0.000	0.000	0.950	0.077	0.995	0.768	0.980	0.364
100	0.000	0.000	0.966	0.173	0.999	0.936	0.990	0.605

Table S2: Coverage of nominal 95% t_{K-p} intervals and SE/SD for β_x , centralized Fed-GEE and decentralized Fed-GEE with $L = 50$ consensus rounds, by scenario. Uncorrected site sandwich throughout.

K	ρ_H	Prev.	p	Coverage					SE/SD				
				Cen.	Comp.	Reg.	Ring	Hub	Cen.	Comp.	Reg.	Ring	Hub
5	0.0	0.30	2	0.953	0.953	0.953	0.953	0.953	0.791	0.791	0.791	0.791	0.791
5	0.0	0.07	2	0.940	0.940	0.940	0.940	0.940	0.820	0.820	0.820	0.820	0.820
5	0.1	0.30	2	0.930	0.930	0.930	0.930	0.930	0.762	0.762	0.762	0.762	0.762
5	0.1	0.07	2	0.925	0.925	0.925	0.925	0.925	0.753	0.753	0.753	0.753	0.753
5	0.2	0.30	2	0.940	0.940	0.940	0.940	0.940	0.778	0.778	0.778	0.778	0.778
5	0.2	0.07	2	0.925	0.925	0.925	0.925	0.925	0.756	0.756	0.756	0.756	0.756

K	ρ_H	Prev.	p	Cen.	Comp.	Reg.	Ring	Hub	Cen.	Comp.	Reg.	Ring	Hub
10	0.0	0.30	2	0.940	0.940	0.940	0.940	0.940	0.909	0.909	0.909	0.909	0.909
10	0.0	0.07	2	0.927	0.927	0.927	0.927	0.927	0.876	0.876	0.876	0.876	0.876
10	0.1	0.30	2	0.945	0.945	0.945	0.945	0.945	0.931	0.931	0.931	0.931	0.931
10	0.1	0.07	2	0.920	0.920	0.920	0.920	0.920	0.846	0.846	0.846	0.846	0.846
10	0.2	0.30	2	0.900	0.900	0.900	0.900	0.900	0.789	0.789	0.789	0.789	0.789
10	0.2	0.07	2	0.930	0.930	0.930	0.930	0.930	0.880	0.880	0.879	0.880	0.880
15	0.0	0.30	2	0.935	0.935	0.935	0.935	0.935	0.952	0.952	0.952	0.952	0.951
15	0.0	0.07	2	0.953	0.953	0.953	0.953	0.953	1.006	1.006	1.006	1.006	1.005
15	0.1	0.30	2	0.935	0.935	0.935	0.935	0.935	0.894	0.894	0.894	0.894	0.894
15	0.1	0.07	2	0.902	0.902	0.902	0.902	0.902	0.886	0.886	0.885	0.886	0.885
15	0.2	0.30	2	0.965	0.965	0.965	0.965	0.965	0.998	0.998	0.998	0.998	0.998
15	0.2	0.07	2	0.935	0.935	0.935	0.935	0.935	0.839	0.839	0.839	0.839	0.838
15	0.0	0.30	3	0.950	0.950	0.950	0.950	0.950	0.968	0.968	0.968	0.968	0.968
15	0.1	0.30	3	0.945	0.945	0.945	0.945	0.945	0.918	0.918	0.918	0.918	0.917
15	0.2	0.30	3	0.920	0.920	0.920	0.920	0.920	0.908	0.908	0.908	0.908	0.907
25	0.0	0.30	2	0.943	0.943	0.943	0.938	0.940	0.930	0.930	0.930	0.915	0.913
25	0.0	0.07	2	0.935	0.935	0.935	0.935	0.932	0.897	0.897	0.896	0.884	0.881
25	0.1	0.30	2	0.955	0.955	0.955	0.953	0.950	1.006	1.006	1.005	0.992	0.988
25	0.1	0.07	2	0.945	0.945	0.945	0.940	0.943	0.965	0.965	0.964	0.952	0.946
25	0.2	0.30	2	0.970	0.970	0.970	0.970	0.960	0.995	0.995	0.995	0.980	0.978
25	0.2	0.07	2	0.925	0.925	0.925	0.920	0.920	0.942	0.942	0.942	0.924	0.924
25	0.0	0.30	3	0.960	0.960	0.960	0.958	0.958	1.002	1.002	1.001	0.993	0.982
25	0.1	0.30	3	0.955	0.955	0.955	0.950	0.953	0.992	0.992	0.992	0.981	0.976
25	0.2	0.30	3	0.915	0.915	0.915	0.905	0.910	0.845	0.845	0.845	0.831	0.828
50	0.0	0.30	2	0.958	0.958	0.958	0.945	0.912	1.012	1.012	1.011	0.984	0.838
50	0.0	0.07	2	0.953	0.953	0.953	0.948	0.895	1.011	1.011	1.010	0.982	0.836
50	0.1	0.30	2	0.927	0.927	0.930	0.930	0.885	0.990	0.990	0.989	0.958	0.829
50	0.1	0.07	2	0.935	0.935	0.935	0.920	0.877	0.922	0.922	0.921	0.889	0.758
50	0.2	0.30	2	0.920	0.920	0.920	0.920	0.865	0.926	0.926	0.924	0.883	0.772
50	0.2	0.07	2	0.945	0.945	0.945	0.925	0.840	0.946	0.946	0.945	0.897	0.790
50	0.0	0.30	3	0.948	0.948	0.948	0.943	0.885	0.996	0.996	0.995	0.970	0.825
50	0.1	0.30	3	0.927	0.927	0.927	0.912	0.850	0.943	0.943	0.942	0.907	0.790
50	0.2	0.30	3	0.935	0.935	0.935	0.870	0.795	0.944	0.944	0.943	0.894	0.800
100	0.0	0.30	2	0.935	0.935	0.935	0.925	0.562	0.970	0.970	0.962	0.928	0.514
100	0.0	0.07	2	0.960	0.960	0.958	0.953	0.555	1.015	1.015	1.008	0.974	0.521
100	0.1	0.30	2	0.958	0.958	0.955	0.940	0.362	1.011	1.011	1.004	0.970	0.537
100	0.1	0.07	2	0.950	0.950	0.948	0.922	0.482	0.997	0.997	0.990	0.950	0.553
100	0.2	0.30	2	0.955	0.955	0.955	0.940	0.200	0.988	0.988	0.981	0.940	0.563

K	ρ_H	Prev.	p	Cen.	Comp.	Reg.	Ring	Hub	Cen.	Comp.	Reg.	Ring	Hub
100	0.2	0.07	2	0.925	0.925	0.915	0.905	0.335	0.969	0.969	0.961	0.932	0.558
100	0.0	0.30	3	0.945	0.945	0.943	0.930	0.580	0.997	0.997	0.991	0.961	0.527
100	0.1	0.30	3	0.940	0.940	0.940	0.912	0.265	0.989	0.989	0.983	0.950	0.545
100	0.2	0.30	3	0.965	0.965	0.965	0.920	0.060	1.047	1.047	1.041	1.026	0.630

J Full simulation results

Tables S3 and S4 report every scenario of the simulation study in the main text, without averaging. Prev. is the average outcome prevalence (0.07, 0.30, and 0.50 for $\beta_0 = -3, -1, 0$). CV is the coefficient of variation of site size. t_{BM} is the t reference on score-space Bell–McCaffrey degrees of freedom. t_{FG} uses the degrees of freedom of Fay and Graubard. t_ν uses the effective-contributor count $(\sum_i q_{ij})^2 / \sum_i q_{ij}^2$, where q_{ij} is site i ’s contribution to the corrected variance of coefficient j . It is shown for comparison. Its coverage was at least 0.95 in every scenario, because it uses the realized scores and understates the degrees of freedom. ν^{BM} and ν^{FG} are median degrees of freedom. All 117,000 replicates completed.

Table S3: Coverage of nominal 95% intervals for β_x in all 117 scenarios. Each scenario had 1,000 replicates; Monte Carlo standard error about 0.007.

K	p	Prev.	CV	Uncorrected		KC			FG		MD	KC	ν^{BM}	ν^{FG}
				z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_ν		
10	2	0.50	0.30	0.883	0.927	0.937	0.941	0.942	0.942	0.943	0.950	0.965	8.1	8.0
10	2	0.50	0.78	0.876	0.932	0.933	0.934	0.944	0.944	0.950	0.963	0.966	5.9	5.4
10	2	0.50	1.00	0.876	0.935	0.930	0.934	0.958	0.957	0.964	0.976	0.966	5.0	4.5
10	2	0.30	0.30	0.909	0.943	0.950	0.951	0.952	0.952	0.953	0.963	0.975	8.1	8.0
10	2	0.30	0.78	0.877	0.939	0.938	0.941	0.956	0.956	0.961	0.970	0.972	5.9	5.4
10	2	0.30	1.00	0.875	0.938	0.931	0.936	0.955	0.955	0.961	0.972	0.964	5.0	4.5
10	2	0.07	0.30	0.882	0.934	0.942	0.944	0.943	0.944	0.946	0.960	0.975	7.9	7.6
10	2	0.07	0.78	0.874	0.940	0.935	0.948	0.959	0.960	0.966	0.976	0.981	5.7	5.3
10	2	0.07	1.00	0.867	0.949	0.941	0.945	0.965	0.965	0.970	0.977	0.973	4.9	4.4
10	5	0.50	0.30	0.903	0.938	0.941	0.964	0.947	0.944	0.944	0.956	0.970	8.1	8.0
10	5	0.50	0.78	0.892	0.938	0.938	0.968	0.959	0.959	0.965	0.975	0.978	5.9	5.4
10	5	0.50	1.00	0.881	0.945	0.943	0.966	0.969	0.970	0.974	0.983	0.969	5.0	4.4
10	5	0.30	0.30	0.909	0.939	0.950	0.973	0.954	0.955	0.956	0.966	0.981	8.1	7.9
10	5	0.30	0.78	0.876	0.933	0.934	0.957	0.951	0.947	0.951	0.962	0.969	5.8	5.4
10	5	0.30	1.00	0.876	0.947	0.947	0.965	0.967	0.967	0.971	0.978	0.971	5.0	4.4
10	5	0.07	0.30	0.893	0.936	0.939	0.969	0.947	0.949	0.948	0.962	0.974	7.7	7.5

K	p	Prev.	CV	z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_{ν}	ν^{BM}	ν^{FG}
10	5	0.07	0.78	0.882	0.941	0.936	0.963	0.962	0.964	0.964	0.972	0.973	5.7	5.1
10	5	0.07	1.00	0.863	0.929	0.924	0.954	0.954	0.959	0.967	0.974	0.971	4.9	4.3
18	2	0.50	0.30	0.922	0.943	0.947	0.947	0.947	0.948	0.948	0.955	0.973	15.3	15.2
18	2	0.50	0.78	0.892	0.930	0.932	0.933	0.943	0.942	0.945	0.951	0.959	10.9	10.3
18	2	0.50	1.00	0.925	0.948	0.948	0.948	0.963	0.960	0.967	0.974	0.974	9.0	8.3
18	2	0.30	0.30	0.917	0.937	0.945	0.946	0.949	0.949	0.949	0.959	0.975	15.3	15.1
18	2	0.30	0.78	0.900	0.939	0.939	0.941	0.951	0.951	0.953	0.959	0.968	10.9	10.3
18	2	0.30	1.00	0.918	0.942	0.939	0.939	0.952	0.952	0.954	0.964	0.967	9.0	8.3
18	2	0.07	0.30	0.924	0.941	0.945	0.946	0.949	0.950	0.950	0.958	0.971	14.7	14.5
18	2	0.07	0.78	0.911	0.942	0.938	0.940	0.951	0.952	0.957	0.960	0.969	10.6	10.0
18	2	0.07	1.00	0.909	0.943	0.940	0.941	0.951	0.951	0.955	0.962	0.964	8.9	8.2
18	5	0.50	0.30	0.942	0.953	0.955	0.956	0.956	0.956	0.956	0.960	0.976	15.3	15.2
18	5	0.50	0.78	0.920	0.946	0.946	0.954	0.957	0.958	0.959	0.966	0.971	10.9	10.3
18	5	0.50	1.00	0.903	0.931	0.931	0.935	0.945	0.945	0.949	0.954	0.961	9.0	8.4
18	5	0.30	0.30	0.922	0.945	0.955	0.961	0.957	0.957	0.957	0.965	0.976	15.3	15.1
18	5	0.30	0.78	0.912	0.942	0.946	0.951	0.955	0.955	0.956	0.968	0.971	10.9	10.2
18	5	0.30	1.00	0.891	0.929	0.928	0.930	0.935	0.939	0.944	0.955	0.965	8.9	8.3
18	5	0.07	0.30	0.912	0.930	0.939	0.947	0.943	0.944	0.944	0.952	0.971	14.7	14.4
18	5	0.07	0.78	0.905	0.937	0.938	0.943	0.945	0.948	0.950	0.956	0.971	10.6	9.9
18	5	0.07	1.00	0.897	0.922	0.920	0.924	0.939	0.941	0.944	0.958	0.959	8.8	7.9
18	10	0.50	0.30	0.928	0.956	0.959	0.971	0.959	0.959	0.959	0.963	0.977	15.3	15.2
18	10	0.50	0.78	0.920	0.941	0.942	0.952	0.948	0.948	0.949	0.955	0.962	10.7	10.2
18	10	0.50	1.00	0.906	0.938	0.934	0.952	0.953	0.952	0.955	0.960	0.966	9.0	8.4
18	10	0.30	0.30	0.933	0.952	0.956	0.970	0.958	0.958	0.958	0.965	0.975	15.3	15.1
18	10	0.30	0.78	0.921	0.950	0.952	0.966	0.957	0.956	0.960	0.967	0.975	10.9	10.4
18	10	0.30	1.00	0.902	0.933	0.935	0.952	0.949	0.946	0.949	0.963	0.964	9.0	8.4
18	10	0.07	0.30	0.920	0.954	0.958	0.970	0.959	0.959	0.959	0.969	0.974	14.6	14.3
18	10	0.07	0.78	0.895	0.928	0.929	0.948	0.943	0.944	0.944	0.955	0.961	10.4	9.8
18	10	0.07	1.00	0.876	0.920	0.914	0.935	0.935	0.938	0.946	0.955	0.954	8.8	8.0
30	2	0.50	0.30	0.936	0.947	0.950	0.951	0.952	0.952	0.952	0.956	0.970	26.2	26.0
30	2	0.50	0.78	0.935	0.951	0.952	0.952	0.956	0.956	0.956	0.960	0.968	18.2	17.6
30	2	0.50	1.00	0.926	0.944	0.944	0.945	0.951	0.951	0.951	0.958	0.963	14.9	14.2
30	2	0.30	0.30	0.930	0.944	0.946	0.946	0.946	0.947	0.947	0.953	0.970	26.1	25.9
30	2	0.30	0.78	0.924	0.937	0.937	0.937	0.943	0.943	0.943	0.949	0.960	18.3	17.6
30	2	0.30	1.00	0.918	0.939	0.936	0.936	0.947	0.947	0.947	0.953	0.962	15.0	14.2
30	2	0.07	0.30	0.938	0.946	0.950	0.951	0.953	0.953	0.953	0.957	0.968	25.2	24.9
30	2	0.07	0.78	0.932	0.940	0.942	0.943	0.949	0.949	0.950	0.961	0.967	17.7	17.1
30	2	0.07	1.00	0.920	0.942	0.939	0.940	0.949	0.949	0.950	0.956	0.964	14.7	13.9
30	5	0.50	0.30	0.932	0.947	0.948	0.950	0.949	0.950	0.950	0.951	0.960	26.2	26.0
30	5	0.50	0.78	0.925	0.940	0.942	0.944	0.948	0.948	0.948	0.950	0.963	18.3	17.6

K	p	Prev.	CV	z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_{ν}	ν^{BM}	ν^{FG}
30	5	0.50	1.00	0.910	0.936	0.936	0.937	0.947	0.947	0.947	0.952	0.954	15.2	14.4
30	5	0.30	0.30	0.928	0.935	0.940	0.941	0.940	0.940	0.940	0.945	0.953	26.0	25.9
30	5	0.30	0.78	0.910	0.930	0.929	0.932	0.939	0.939	0.941	0.946	0.962	18.2	17.5
30	5	0.30	1.00	0.917	0.938	0.937	0.939	0.946	0.946	0.947	0.955	0.963	14.9	14.1
30	5	0.07	0.30	0.913	0.931	0.936	0.938	0.938	0.939	0.939	0.944	0.963	25.0	24.7
30	5	0.07	0.78	0.910	0.925	0.924	0.926	0.932	0.931	0.935	0.945	0.965	17.8	17.1
30	5	0.07	1.00	0.927	0.945	0.943	0.944	0.949	0.948	0.948	0.954	0.965	14.6	13.7
30	10	0.50	0.30	0.933	0.942	0.943	0.947	0.944	0.943	0.943	0.949	0.960	26.1	26.0
30	10	0.50	0.78	0.924	0.944	0.944	0.951	0.952	0.952	0.952	0.960	0.971	18.5	17.9
30	10	0.50	1.00	0.927	0.950	0.949	0.951	0.957	0.957	0.959	0.967	0.976	15.1	14.3
30	10	0.30	0.30	0.930	0.939	0.942	0.948	0.942	0.942	0.942	0.950	0.964	26.1	25.9
30	10	0.30	0.78	0.925	0.937	0.942	0.948	0.951	0.949	0.951	0.959	0.970	18.3	17.6
30	10	0.30	1.00	0.924	0.940	0.941	0.942	0.947	0.949	0.951	0.959	0.963	15.2	14.4
30	10	0.07	0.30	0.924	0.934	0.941	0.943	0.943	0.943	0.943	0.946	0.959	25.1	24.8
30	10	0.07	0.78	0.926	0.937	0.937	0.941	0.942	0.943	0.945	0.950	0.958	17.6	16.9
30	10	0.07	1.00	0.929	0.949	0.947	0.953	0.957	0.955	0.956	0.963	0.965	14.7	13.8
30	20	0.50	0.30	0.913	0.935	0.938	0.958	0.939	0.938	0.938	0.943	0.954	26.1	26.0
30	20	0.50	0.78	0.927	0.941	0.942	0.964	0.949	0.948	0.950	0.960	0.965	18.2	17.6
30	20	0.50	1.00	0.913	0.934	0.933	0.948	0.938	0.939	0.939	0.949	0.953	15.0	14.2
30	20	0.30	0.30	0.919	0.938	0.942	0.965	0.944	0.944	0.944	0.949	0.966	26.1	25.9
30	20	0.30	0.78	0.935	0.956	0.954	0.973	0.964	0.961	0.962	0.970	0.972	18.2	17.6
30	20	0.30	1.00	0.927	0.953	0.952	0.964	0.960	0.960	0.960	0.964	0.971	15.0	14.2
30	20	0.07	0.30	0.918	0.927	0.929	0.952	0.932	0.930	0.930	0.938	0.951	25.1	24.7
30	20	0.07	0.78	0.925	0.940	0.942	0.956	0.947	0.946	0.946	0.954	0.963	17.6	16.9
30	20	0.07	1.00	0.913	0.930	0.931	0.947	0.941	0.944	0.948	0.954	0.952	14.6	13.7
50	2	0.50	0.30	0.937	0.944	0.946	0.946	0.946	0.946	0.946	0.950	0.957	44.2	44.0
50	2	0.50	0.78	0.952	0.957	0.958	0.958	0.963	0.963	0.963	0.966	0.969	30.7	30.0
50	2	0.50	1.00	0.946	0.955	0.954	0.955	0.962	0.962	0.963	0.966	0.969	24.9	24.1
50	2	0.30	0.30	0.941	0.949	0.951	0.951	0.952	0.952	0.952	0.953	0.959	44.1	43.9
50	2	0.30	0.78	0.942	0.953	0.952	0.952	0.955	0.955	0.955	0.958	0.963	30.5	29.8
50	2	0.30	1.00	0.932	0.945	0.944	0.944	0.952	0.952	0.952	0.957	0.969	24.9	24.0
50	2	0.07	0.30	0.934	0.939	0.941	0.941	0.941	0.941	0.941	0.942	0.954	42.5	42.3
50	2	0.07	0.78	0.912	0.924	0.925	0.926	0.929	0.929	0.930	0.941	0.956	29.7	28.9
50	2	0.07	1.00	0.939	0.951	0.950	0.950	0.955	0.956	0.957	0.961	0.967	24.5	23.6
50	5	0.50	0.30	0.940	0.946	0.950	0.950	0.950	0.950	0.950	0.952	0.960	44.2	44.0
50	5	0.50	0.78	0.947	0.953	0.954	0.955	0.959	0.958	0.959	0.960	0.969	30.6	30.0
50	5	0.50	1.00	0.933	0.943	0.944	0.944	0.951	0.950	0.952	0.958	0.971	25.0	24.1
50	5	0.30	0.30	0.947	0.953	0.954	0.955	0.955	0.955	0.955	0.956	0.961	44.0	43.8
50	5	0.30	0.78	0.945	0.962	0.965	0.965	0.971	0.970	0.970	0.975	0.980	30.6	30.0
50	5	0.30	1.00	0.918	0.930	0.929	0.931	0.941	0.941	0.941	0.947	0.958	25.0	24.2

K	p	Prev.	CV	z	t_{BM}	t_{K-1}	t_{K-p}	t_{BM}	t_{BM}	t_{FG}	t_{BM}	t_{ν}	ν^{BM}	ν^{FG}
50	5	0.07	0.30	0.945	0.948	0.952	0.953	0.953	0.953	0.953	0.955	0.967	42.4	42.1
50	5	0.07	0.78	0.928	0.936	0.936	0.937	0.943	0.942	0.942	0.946	0.960	29.7	28.9
50	5	0.07	1.00	0.929	0.940	0.940	0.940	0.945	0.944	0.944	0.951	0.960	24.2	23.3
50	10	0.50	0.30	0.941	0.949	0.949	0.949	0.949	0.949	0.949	0.952	0.959	44.2	44.0
50	10	0.50	0.78	0.940	0.950	0.951	0.952	0.953	0.953	0.953	0.956	0.959	30.7	30.0
50	10	0.50	1.00	0.928	0.940	0.940	0.940	0.944	0.944	0.944	0.951	0.961	25.2	24.3
50	10	0.30	0.30	0.944	0.949	0.950	0.950	0.950	0.950	0.950	0.953	0.963	44.0	43.8
50	10	0.30	0.78	0.932	0.937	0.938	0.938	0.940	0.940	0.940	0.946	0.954	30.6	29.9
50	10	0.30	1.00	0.938	0.947	0.947	0.947	0.949	0.950	0.950	0.953	0.965	25.0	24.1
50	10	0.07	0.30	0.942	0.950	0.953	0.954	0.954	0.954	0.954	0.956	0.968	42.4	42.2
50	10	0.07	0.78	0.944	0.952	0.952	0.953	0.955	0.955	0.956	0.961	0.973	29.8	29.0
50	10	0.07	1.00	0.928	0.937	0.937	0.937	0.941	0.943	0.944	0.948	0.960	24.5	23.4
50	20	0.50	0.30	0.942	0.950	0.953	0.955	0.953	0.953	0.953	0.955	0.960	44.2	44.0
50	20	0.50	0.78	0.944	0.952	0.952	0.953	0.954	0.954	0.954	0.954	0.962	30.4	29.7
50	20	0.50	1.00	0.914	0.927	0.928	0.937	0.939	0.938	0.939	0.946	0.957	25.2	24.3
50	20	0.30	0.30	0.940	0.944	0.946	0.950	0.946	0.946	0.946	0.950	0.953	44.0	43.8
50	20	0.30	0.78	0.932	0.943	0.944	0.946	0.949	0.947	0.948	0.949	0.959	30.5	29.8
50	20	0.30	1.00	0.934	0.945	0.944	0.946	0.949	0.949	0.949	0.952	0.956	25.1	24.2
50	20	0.07	0.30	0.945	0.951	0.953	0.958	0.954	0.954	0.954	0.958	0.963	42.3	41.9
50	20	0.07	0.78	0.927	0.938	0.938	0.944	0.944	0.944	0.944	0.949	0.958	29.6	28.9
50	20	0.07	1.00	0.924	0.941	0.940	0.942	0.943	0.942	0.942	0.952	0.959	24.2	23.3

Table S4: Standard-error calibration (SE/SD) by variance estimator, and Fed-GEE (KC with t_{BM}) compared with meta-analytic GEE, in all 117 scenarios.

K	p	Prev.	CV	SE/SD				Fed-GEE		Meta-analytic GEE	
				Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
10	2	0.50	0.30	0.894	0.949	1.008	0.947	0.942	0.10	0.912	0.23
10	2	0.50	0.78	0.860	0.932	1.016	0.931	0.944	0.12	0.893	0.17
10	2	0.50	1.00	0.862	0.946	1.049	0.945	0.958	0.08	0.890	0.18
10	2	0.30	0.30	0.918	0.975	1.036	0.974	0.952	0.11	0.909	0.19
10	2	0.30	0.78	0.854	0.927	1.012	0.926	0.956	0.07	0.875	0.07
10	2	0.30	1.00	0.813	0.894	0.991	0.893	0.955	0.07	0.854	0.07
10	2	0.07	0.30	0.865	0.920	0.980	0.923	0.943	0.12	0.726	-0.31
10	2	0.07	0.78	0.846	0.920	1.005	0.924	0.959	0.13	0.723	-0.25
10	2	0.07	1.00	0.822	0.907	1.012	0.912	0.965	0.12	0.733	-0.22
10	5	0.50	0.30	0.924	0.983	1.048	0.980	0.947	0.22	0.876	0.51
10	5	0.50	0.78	0.882	0.958	1.047	0.955	0.959	0.18	0.884	0.39

K	p	Prev.	CV	Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
10	5	0.50	1.00	0.880	0.969	1.078	0.967	0.969	0.16	0.882	0.39
10	5	0.30	0.30	0.928	0.988	1.054	0.987	0.954	0.16	0.892	0.36
10	5	0.30	0.78	0.852	0.928	1.016	0.928	0.951	0.16	0.870	0.33
10	5	0.30	1.00	0.866	0.956	1.068	0.955	0.967	0.21	0.867	0.36
10	5	0.07	0.30	0.890	0.951	1.020	0.952	0.947	0.13	0.787	-0.12
10	5	0.07	0.78	0.855	0.934	1.028	0.939	0.962	0.12	0.818	-0.06
10	5	0.07	1.00	0.824	0.913	1.024	0.920	0.954	0.17	0.818	0.02
18	2	0.50	0.30	0.940	0.972	1.004	0.970	0.947	0.10	0.888	0.27
18	2	0.50	0.78	0.870	0.910	0.953	0.909	0.943	0.08	0.885	0.20
18	2	0.50	1.00	0.953	1.006	1.066	1.005	0.963	0.09	0.865	0.19
18	2	0.30	0.30	0.927	0.958	0.990	0.957	0.949	0.05	0.900	0.16
18	2	0.30	0.78	0.908	0.950	0.996	0.949	0.951	0.08	0.873	0.06
18	2	0.30	1.00	0.941	0.994	1.053	0.994	0.952	0.10	0.859	0.05
18	2	0.07	0.30	0.938	0.970	1.004	0.972	0.949	0.07	0.641	-0.54
18	2	0.07	0.78	0.906	0.949	0.996	0.951	0.951	0.03	0.654	-0.52
18	2	0.07	1.00	0.893	0.944	1.000	0.945	0.951	0.09	0.686	-0.44
18	5	0.50	0.30	1.002	1.037	1.073	1.034	0.956	0.14	0.876	0.60
18	5	0.50	0.78	0.947	0.993	1.043	0.990	0.957	0.14	0.867	0.50
18	5	0.50	1.00	0.874	0.924	0.980	0.923	0.945	0.15	0.836	0.46
18	5	0.30	0.30	1.007	1.042	1.078	1.040	0.957	0.11	0.887	0.48
18	5	0.30	0.78	0.946	0.992	1.041	0.991	0.955	0.18	0.856	0.46
18	5	0.30	1.00	0.894	0.946	1.005	0.947	0.935	0.18	0.849	0.41
18	5	0.07	0.30	0.918	0.952	0.988	0.951	0.943	0.12	0.748	-0.22
18	5	0.07	0.78	0.923	0.970	1.022	0.972	0.945	0.13	0.803	-0.15
18	5	0.07	1.00	0.871	0.922	0.981	0.927	0.939	0.11	0.817	-0.13
18	10	0.50	0.30	0.978	1.014	1.051	1.009	0.959	0.19	0.618	1.37
18	10	0.50	0.78	0.916	0.961	1.011	0.957	0.948	0.18	0.699	0.99
18	10	0.50	1.00	0.918	0.972	1.033	0.969	0.953	0.22	0.735	1.00
18	10	0.30	0.30	0.976	1.012	1.049	1.009	0.958	0.19	0.685	1.22
18	10	0.30	0.78	0.982	1.031	1.085	1.027	0.957	0.23	0.757	1.03
18	10	0.30	1.00	0.888	0.939	0.998	0.937	0.949	0.19	0.757	0.86
18	10	0.07	0.30	0.941	0.979	1.020	0.976	0.959	0.16	0.800	0.29
18	10	0.07	0.78	0.884	0.933	0.986	0.931	0.943	0.13	0.845	0.31
18	10	0.07	1.00	0.852	0.905	0.966	0.910	0.935	0.21	0.844	0.29
30	2	0.50	0.30	0.996	1.016	1.036	1.015	0.952	-0.01	0.912	0.28
30	2	0.50	0.78	0.973	1.000	1.029	1.000	0.956	0.08	0.883	0.25
30	2	0.50	1.00	0.940	0.972	1.007	0.971	0.951	0.07	0.855	0.19
30	2	0.30	0.30	0.956	0.975	0.994	0.975	0.946	0.06	0.895	0.21
30	2	0.30	0.78	0.917	0.943	0.970	0.942	0.943	0.06	0.862	0.07
30	2	0.30	1.00	0.935	0.968	1.002	0.967	0.947	0.05	0.851	0.04

K	p	Prev.	CV	Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
30	2	0.07	0.30	0.970	0.989	1.009	0.990	0.953	0.04	0.543	-0.76
30	2	0.07	0.78	0.925	0.951	0.979	0.953	0.949	0.06	0.592	-0.63
30	2	0.07	1.00	0.920	0.951	0.985	0.953	0.949	0.09	0.633	-0.59
30	5	0.50	0.30	0.959	0.978	0.999	0.977	0.949	0.07	0.841	0.69
30	5	0.50	0.78	0.947	0.974	1.003	0.973	0.948	0.06	0.836	0.55
30	5	0.50	1.00	0.884	0.915	0.948	0.913	0.947	0.15	0.779	0.57
30	5	0.30	0.30	0.943	0.963	0.983	0.961	0.940	0.12	0.842	0.58
30	5	0.30	0.78	0.951	0.978	1.007	0.977	0.939	0.14	0.844	0.52
30	5	0.30	1.00	0.939	0.972	1.007	0.970	0.946	0.10	0.831	0.43
30	5	0.07	0.30	0.961	0.982	1.004	0.981	0.938	0.04	0.742	-0.38
30	5	0.07	0.78	0.910	0.937	0.966	0.938	0.932	0.08	0.770	-0.24
30	5	0.07	1.00	0.924	0.958	0.995	0.959	0.949	0.04	0.760	-0.27
30	10	0.50	0.30	0.971	0.992	1.013	0.990	0.944	0.11	0.489	1.64
30	10	0.50	0.78	0.952	0.980	1.009	0.977	0.952	0.18	0.580	1.42
30	10	0.50	1.00	0.982	1.018	1.057	1.015	0.957	0.16	0.634	1.31
30	10	0.30	0.30	0.972	0.992	1.014	0.990	0.942	0.14	0.531	1.58
30	10	0.30	0.78	0.941	0.970	1.000	0.967	0.951	0.15	0.621	1.21
30	10	0.30	1.00	0.973	1.007	1.044	1.006	0.947	0.20	0.655	1.14
30	10	0.07	0.30	0.940	0.963	0.986	0.961	0.943	0.15	0.792	0.31
30	10	0.07	0.78	0.926	0.956	0.988	0.954	0.942	0.08	0.820	0.28
30	10	0.07	1.00	0.954	0.990	1.028	0.990	0.957	0.15	0.853	0.30
30	20	0.50	0.30	0.924	0.945	0.967	0.941	0.939	0.23	0.066	2.83
30	20	0.50	0.78	0.935	0.964	0.996	0.960	0.949	0.20	0.227	2.18
30	20	0.50	1.00	0.900	0.933	0.969	0.929	0.938	0.29	0.283	2.04
30	20	0.30	0.30	0.955	0.978	1.001	0.974	0.944	0.13	0.135	2.49
30	20	0.30	0.78	0.962	0.992	1.025	0.988	0.964	0.20	0.297	2.02
30	20	0.30	1.00	0.941	0.977	1.015	0.974	0.960	0.23	0.385	1.81
30	20	0.07	0.30	0.923	0.948	0.975	0.943	0.932	0.19	0.805	0.59
30	20	0.07	0.78	0.937	0.971	1.007	0.965	0.947	0.18	0.776	0.72
30	20	0.07	1.00	0.909	0.947	0.990	0.943	0.941	0.24	0.786	0.72
50	2	0.50	0.30	0.968	0.979	0.990	0.979	0.946	0.01	0.893	0.31
50	2	0.50	0.78	0.990	1.006	1.023	1.006	0.963	0.02	0.891	0.25
50	2	0.50	1.00	0.977	0.997	1.018	0.996	0.962	0.08	0.877	0.24
50	2	0.30	0.30	0.969	0.981	0.992	0.981	0.952	0.02	0.911	0.24
50	2	0.30	0.78	0.981	0.998	1.015	0.997	0.955	0.06	0.870	0.12
50	2	0.30	1.00	0.954	0.973	0.994	0.973	0.952	0.05	0.844	0.03
50	2	0.07	0.30	0.951	0.962	0.974	0.963	0.941	0.02	0.433	-1.07
50	2	0.07	0.78	0.926	0.942	0.958	0.943	0.929	0.06	0.507	-0.89
50	2	0.07	1.00	0.981	1.001	1.023	1.002	0.955	0.08	0.538	-0.85
50	5	0.50	0.30	1.014	1.027	1.039	1.026	0.950	-0.01	0.822	0.82

K	p	Prev.	CV	Unc.	KC	MD	FG	Cov.	Bias/SD	Cov.	Bias/SD
50	5	0.50	0.78	0.996	1.013	1.031	1.012	0.959	0.06	0.775	0.77
50	5	0.50	1.00	0.953	0.973	0.994	0.972	0.951	0.09	0.763	0.73
50	5	0.30	0.30	0.999	1.011	1.024	1.010	0.955	0.10	0.813	0.77
50	5	0.30	0.78	1.016	1.034	1.052	1.033	0.971	0.11	0.795	0.70
50	5	0.30	1.00	0.922	0.942	0.962	0.941	0.941	0.13	0.775	0.52
50	5	0.07	0.30	1.016	1.029	1.043	1.029	0.953	0.06	0.692	-0.48
50	5	0.07	0.78	0.928	0.945	0.962	0.945	0.943	0.06	0.733	-0.37
50	5	0.07	1.00	0.941	0.962	0.984	0.962	0.945	0.06	0.753	-0.35
50	10	0.50	0.30	0.976	0.988	1.001	0.987	0.949	0.12	0.301	2.11
50	10	0.50	0.78	0.967	0.984	1.002	0.982	0.953	0.17	0.414	1.75
50	10	0.50	1.00	0.953	0.974	0.995	0.972	0.944	0.20	0.473	1.65
50	10	0.30	0.30	0.974	0.987	1.000	0.986	0.950	0.07	0.396	1.88
50	10	0.30	0.78	0.940	0.957	0.975	0.955	0.940	0.15	0.482	1.57
50	10	0.30	1.00	0.958	0.979	1.001	0.978	0.949	0.14	0.559	1.42
50	10	0.07	0.30	0.991	1.005	1.019	1.004	0.954	0.10	0.787	0.36
50	10	0.07	0.78	0.980	0.999	1.018	0.998	0.955	0.08	0.822	0.35
50	10	0.07	1.00	0.941	0.962	0.985	0.962	0.941	0.11	0.817	0.40
50	20	0.50	0.30	0.985	0.998	1.012	0.996	0.953	0.16	0.010	3.68
50	20	0.50	0.78	0.976	0.994	1.014	0.992	0.954	0.15	0.073	2.84
50	20	0.50	1.00	0.927	0.947	0.969	0.945	0.939	0.17	0.132	2.50
50	20	0.30	0.30	0.970	0.984	0.997	0.981	0.946	0.18	0.028	3.15
50	20	0.30	0.78	0.977	0.996	1.015	0.993	0.949	0.19	0.110	2.72
50	20	0.30	1.00	0.926	0.947	0.969	0.945	0.949	0.15	0.190	2.32
50	20	0.07	0.30	0.982	0.998	1.015	0.995	0.954	0.13	0.740	0.70
50	20	0.07	0.78	0.945	0.965	0.986	0.962	0.944	0.13	0.727	0.82
50	20	0.07	1.00	0.930	0.953	0.978	0.951	0.943	0.22	0.711	0.92

K VHA application: additional results

Web Table S5 gives the VHA confidence intervals under the uncorrected sandwich with normal and Bell–McCaffrey references, the KC and FG corrections, FG with its own degrees of freedom, and CR2 with Satterthwaite degrees of freedom. The VISNs are nearly equal in size (CV 0.35), so the corrections change the standard errors by at most 12%. Every column gives the same conclusion for every coefficient.

Table S5: VHA application: 95% confidence intervals for the adjusted odds ratios under alternative variance corrections and reference distributions. All columns share the point estimates in Table 3 of the main text.

Covariate	Uncorrected		KC	FG		CR2
	z	t_{BM}	t_{BM}	t_{BM}	t_{FG}	Satterthwaite
<i>Demographics</i>						
Female	1.05, 1.41	1.04, 1.43	1.03, 1.44	1.03, 1.43	1.03, 1.43	1.03, 1.44
Age ≥ 65 years	0.79, 0.98	0.79, 0.99	0.78, 0.99	0.78, 0.99	0.78, 1.00	0.78, 0.99
<i>Alcohol severity</i>						
AUDIT-C no or low use	0.84, 0.93	0.84, 0.94	0.83, 0.94	0.84, 0.94	0.83, 0.94	0.83, 0.94
Opioid use disorder	0.54, 0.68	0.53, 0.69	0.53, 0.69	0.53, 0.69	0.53, 0.69	0.53, 0.69
<i>Medical complexity</i>						
CAN score ≥ 60 th percentile	0.65, 0.78	0.65, 0.79	0.64, 0.79	0.64, 0.80	0.64, 0.80	0.64, 0.79
Frailty, mild or worse	0.61, 0.78	0.60, 0.78	0.60, 0.79	0.60, 0.79	0.60, 0.79	0.60, 0.79
Any ICU stay	0.76, 1.02	0.75, 1.04	0.74, 1.05	0.74, 1.04	0.74, 1.04	0.74, 1.05
<i>Hospital process</i>						
Psychiatry discharging service	1.43, 2.09	1.41, 2.13	1.39, 2.15	1.39, 2.15	1.39, 2.15	1.39, 2.15
Addiction consult						
Clinician	1.19, 1.82	1.15, 1.87	1.14, 1.90	1.14, 1.90	1.14, 1.90	1.14, 1.90
Non-clinician	1.36, 2.59	1.29, 2.73	1.24, 2.84	1.24, 2.85	1.23, 2.87	1.24, 2.84
Discharge against advice	0.66, 1.92	0.58, 2.19	0.53, 2.41	0.53, 2.41	0.52, 2.46	0.53, 2.40

Uncorrected, VISN-clustered sandwich. KC, FG, score-space corrections. t_{BM} , t reference on score-space Bell–McCaffrey degrees of freedom; t_{FG} , degrees of freedom of Fay and Graubard. CR2, bias-reduced linearization with Satterthwaite degrees of freedom computed on the pooled data with the clubSandwich package. The KC, t_{BM} column is the recommended procedure. CAN, Care Assessment Need; ICU, intensive care unit; AUDIT-C, Alcohol Use Disorders Identification Test–Consumption. Reference categories: male; age 18–64 years; AUDIT-C unhealthy use or higher; no opioid use disorder; CAN score below the 60th percentile; nonfrail or prefrail; no ICU stay; medical or surgical discharging service; no addiction consult; routine discharge.

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Fed-GEE: Federated Generalized Estimating Equations for Multisite Health Data, with an Application to Veterans' Health

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Short title: Small-site inference for federated GEE

Abstract

Joint analysis of multisite health data increases sample size, population diversity, and generalizability. Yet patient records often cannot be pooled, outcomes are correlated within sites, and sites are often few. We propose federated generalized estimating equations (Fed-GEE) for population-averaged associations. By exchanging site score vectors and information matrices, Fed-GEE reproduces the pooled estimate exactly. With few sites as the independent clusters, however, the sandwich variance is too small and normal critical values are too narrow. We derive federated versions of the small-sample variance corrections and reference distributions used in pooled analyses. We give theoretical justification and a simulation-based recommendation: the Kauermann–Carroll correction with Bell–McCaffrey degrees of freedom. In simulations with 10 to 50 sites, Fed-GEE matched pooled GEE to numerical precision, and the recommended procedure kept 95% interval coverage near nominal (0.929 to 0.971), compared with as low as 0.863 without correction. Meta-analysis of site-specific fits covered as low as 0.010. We applied Fed-GEE to 29,041 hospitalizations of 18,900 Veterans with alcohol use disorder, emulating a federation of 18 Veterans Health Administration regional networks. Clinician and non-clinician addiction consultation were each associated with

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starting alcohol use disorder medication (odds ratios 1.47 and 1.88). These findings match the pooled analysis, yet no patient records need leave their sites. Non-clinician consultation was also five times as common (15.0% versus 2.9%). Where non-clinician staff cost less to hire, expanding this workforce could increase consultation capacity within a fixed budget.

Keywords: sandwich variance estimator; small-sample correction; few clusters; Bell–McCaffrey degrees of freedom; distributed inference; alcohol use disorder; addiction consultation

1 Introduction

Alcohol use disorder (AUD) is more common among US Veterans than in the general adult population. National surveys estimated lifetime prevalences of 42.2% and 29.1%, respectively [1, 2]. Among Veterans, AUD is associated with substantial psychiatric morbidity, suicidal behavior, and higher all-cause mortality [1, 3].

Each hospitalization for AUD offers an opportunity to initiate treatment [4]. Effective medications for AUD (MAUD) are available but remain underused [5]. Hospital addiction consultation services are associated with greater initiation of medications for opioid use disorder [6, 7]. Evidence for AUD is more limited, although recent studies suggest that consultation may also support MAUD initiation [8, 9].

Using national Veterans Health Administration (VHA) data, we examine whether hospital addiction consultation is associated with higher MAUD initiation and whether this association is present for consultation delivered by both clinicians and non-clinicians. In this study, clinicians are physicians and advanced practice clinicians; non-clinicians are social workers, nurses, and peer support staff [9]. The latter can support treatment engagement and coordination, with prescribing remaining the responsibility of an authorized clinician. These associations can inform staffing policy. Evidence confined to clinician consultation would favor prioritizing clinician capacity; evidence for non-clinician consultation would support considering a broader workforce. Where non-clinician staff are more available and less costly to hire, this could expand consultation capacity within a fixed budget.

We analyze 29,041 hospitalizations of 18,900 Veterans with a primary diagnosis of AUD in 2022 or 2023 [9]. The outcome is MAUD initiation during hospitalization or within 7 days of discharge. Hospitalizations are clustered within Veterans and within regions that share health services and practice patterns. Our policy target is the population-averaged association between consultation and MAUD initiation across the VHA. Generalized estimating

equations (GEE) provide a routine way to estimate this marginal association while allowing correlated outcomes [10, 11].

The scientific setting adds a need to limit the movement of sensitive records. Pooling patient-level data creates additional data-transfer and access risks, and AUD diagnoses carry stigma that can itself discourage treatment seeking [12, 13]. These concerns motivate a federated system in which records remain under local control and sites exchange statistical summaries. Federated methods for generalized linear mixed models are available [14], but their conditional regression coefficients do not directly answer our population-averaged question. We therefore require a federated GEE procedure that supports both marginal estimation and inference. Keeping records local reduces the need to share sensitive data; formal protection against disclosure from exchanged summaries requires additional safeguards [15].

The VHA’s regional organization creates a further inferential challenge. Its 18 Veterans Integrated Service Networks (VISNs) are administrative units responsible for regional resources and policy. Shared staffing policies and practice patterns motivate treating each VISN as an independent unit in our analysis. Thus, despite the large number of hospitalizations, inference rests on only 18 clusters. This does not prevent computation of the GEE point estimate, but it can undermine the large-cluster approximations used for inference, whether estimation is centralized or decentralized. With few clusters, the usual sandwich variance tends to be biased downward and normal Wald intervals can undercover [16, 17]. Established remedies include the Kauermann–Carroll (KC), Mancl–DeRouen (MD), and Fay–Graubard (FG) variance corrections and small-sample reference distributions [16–20]. Their conventional formulations use observation-level residuals and cluster leverage matrices that are directly available in a pooled analysis. Applying them in a federation requires expressions that can be evaluated from shared summaries. The reference distribution must likewise be determined from these summaries, with critical values that reflect unequal information across the small number of sites [19, 21].

The contributions of this paper are as follows. We develop Fed-GEE for population-averaged estimation and small-sample inference in federated networks, with implementations both with and without a coordinating center. We express the KC, MD, and FG variance corrections in score space, using site score vectors and information matrices without transferring patient-level records. We also compute coefficient-specific Bell–McCaffrey degrees of freedom from these summaries, enabling federated construction of the t reference distributions used for confidence intervals and hypothesis tests. Both the variance correction and reference distribution are computed in a single post-convergence stage. With a coordinating center, this requires one additional exchange of site summaries; without a center, the summaries are aggregated through neighbor consensus. Under an affine working-score

model with covariance correct up to scale, we show that the KC half-power gives an exactly unbiased variance estimator and is the unique exponent in the proposed family with this property. Extensive simulations examine coverage with 10 to 50 sites and unequal site sizes, and additional simulations assess decentralized fitting with an uncorrected sandwich. The decentralized small-sample correction is developed theoretically and is not evaluated in these simulations. Finally, we apply the procedure to the VHA staffing question. The application uses centrally available records to emulate federation across VISNs and verify the results against a pooled analysis.

Section 2 presents the VHA analysis model, federated GEE, score-space corrections, and reference distributions, followed by algorithms with and without a coordinating center. Section 3 reports the simulation study, and Section 4 presents the VHA application. Section 5 discusses implications and limitations.

2 Methods

2.1 Application model and estimand

For hospitalization t of Veteran j in VISN i , let Y_{ijt} indicate MAUD initiation within the outcome window defined above. We specify the marginal logistic model

$$\text{logit}\{\Pr(Y_{ijt} = 1 \mid C_{ijt}, N_{ijt}, Z_{ijt})\} = \beta_0 + \beta_C C_{ijt} + \beta_N N_{ijt} + Z_{ijt}^\top \boldsymbol{\gamma}, \quad (1)$$

where C_{ijt} and N_{ijt} indicate clinician and non-clinician consultation, respectively, with no consultation as the reference. The nine adjustment covariates in Z_{ijt} account for measured differences in case mix and hospital care. They cover demographics (age at least 65 years and female sex), alcohol severity and co-occurring substance use (no or low AUDIT-C score and opioid use disorder), medical complexity (Care Assessment Need score at or above the 60th percentile, mild or worse frailty, and intensive care unit stay), and hospital processes (psychiatry discharging service and discharge against medical advice) [9]. These groups correspond to those in Figure 1.

We fit the model by GEE with working independence and $p = 12$ coefficients. Hospitalizations are nested within Veterans and Veterans within VISNs; the VISN is both the unit of federation and the independent unit for variance estimation. The target estimands, $\exp(\beta_C)$ and $\exp(\beta_N)$, are adjusted population-averaged odds ratios for each consultation type relative to no consultation. They describe associations, not causal effects. The following formulation extends this model to general outcomes and working covariances.

2.2 Estimating equations and federated fitting

Suppose the data come from K independent sites. Site i holds n_i patient clusters, indexed by $j = 1, \dots, n_i$. Cluster j has a response vector Y_{ij} , a covariate matrix X_{ij} , and marginal mean $\mu_{ij}(\boldsymbol{\beta}) = g^{-1}(X_{ij}\boldsymbol{\beta})$ for a known link g and a p -vector $\boldsymbol{\beta}$. Let $D_{ij} = \partial\mu_{ij}/\partial\boldsymbol{\beta}^\top$, and let V_{ij} be a working covariance matrix [10]. Each site computes a score vector and an information matrix,

$$\mathcal{S}_i(\boldsymbol{\beta}) = \sum_{j=1}^{n_i} D_{ij}^\top V_{ij}^{-1} \{Y_{ij} - \mu_{ij}(\boldsymbol{\beta})\}, \quad \mathcal{B}_i(\boldsymbol{\beta}) = \sum_{j=1}^{n_i} D_{ij}^\top V_{ij}^{-1} D_{ij}. \quad (2)$$

The GEE estimator $\hat{\boldsymbol{\beta}}$ solves $\mathcal{S}(\boldsymbol{\beta}) = \sum_i \mathcal{S}_i(\boldsymbol{\beta}) = 0$. Because the score and information matrix are additive across sites, the estimating equation can be solved using site summaries. At the current value $\boldsymbol{\beta}^{(r)}$, each site sends $\mathcal{S}_i\{\boldsymbol{\beta}^{(r)}\}$ and $\mathcal{B}_i\{\boldsymbol{\beta}^{(r)}\}$ to a coordinating center. The center adds them and takes a Fisher-scoring step,

$$\boldsymbol{\beta}^{(r+1)} = \boldsymbol{\beta}^{(r)} + \mathcal{B}\{\boldsymbol{\beta}^{(r)}\}^{-1} \mathcal{S}\{\boldsymbol{\beta}^{(r)}\}, \quad \mathcal{B} = \sum_{i=1}^K \mathcal{B}_i. \quad (3)$$

It then returns $\boldsymbol{\beta}^{(r+1)}$ to the sites. We stop when $\|\boldsymbol{\beta}^{(r+1)} - \boldsymbol{\beta}^{(r)}\| < 10^{-8}$, with at most 50 iterations. Each message has $p + p(p+1)/2$ summary statistics, and no patient-level record leaves a site. Algorithm 1 summarizes the fitting routine and the one-shot post-convergence correction developed in Section 2.5.

2.3 Large-sample inference

Inference uses the cluster-robust sandwich variance at the site level,

$$\widehat{V}_S = \widehat{\mathcal{B}}^{-1} \widehat{\mathcal{M}} \widehat{\mathcal{B}}^{-1}, \quad \widehat{\mathcal{M}} = \sum_{i=1}^K \widehat{\mathcal{S}}_i \widehat{\mathcal{S}}_i^\top, \quad (4)$$

where $\widehat{\mathcal{S}}_i = \mathcal{S}_i(\hat{\boldsymbol{\beta}})$ and $\widehat{\mathcal{B}} = \sum_i \mathcal{B}_i(\hat{\boldsymbol{\beta}})$. The site is the independent unit because patients within a site share practice patterns, staffing, and local policy. The site-level sandwich accommodates dependence among patients within a site and remains asymptotically valid under working-covariance misspecification, provided the marginal mean is correctly specified and the sites are independent [10]. Suppose the sites are independent, p is fixed, $K^{-1}\widehat{\mathcal{B}} \rightarrow B_0$, and $K^{-1}\widehat{\mathcal{M}} \rightarrow M_0$ in probability. Under the regularity conditions of Liang and Zeger [10], $\hat{\boldsymbol{\beta}}$ is consistent and

$$\sqrt{K}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(0, B_0^{-1} M_0 B_0^{-1}) \quad \text{as } K \rightarrow \infty.$$

Hence $\widehat{V}_S$ estimates $\text{Var}(\widehat{\beta})$ directly, and Wald intervals use normal critical values. Web Appendix A states the conditions. The result is standard, because $\widehat{\beta}$ is the pooled GEE estimator.

2.4 The small-site problem

The large-sample approximation depends on the number of independent sites, which can be small even in a large health-system cohort. Estimating the regression coefficients from these same sites constrains the fitted scores and can bias the sandwich variance downward.

Collect the fitted scores as the columns of $S = [\widehat{S}_1, \dots, \widehat{S}_K] \in \mathbb{R}^{p \times K}$. Since $S\mathbf{1}_K = 0$ at the GEE solution,

$$\text{rank}(\widehat{\mathcal{M}}) \leq \min(p, K - 1). \quad (5)$$

Thus, the meat is singular whenever $K \leq p$ (Web Appendix E). For $K > p$, the bound does not imply singularity, but the variance estimate still depends on at most $K - 1$ independent score contrasts.

Fitting also reduces the expected score variation under a correctly specified affine working model. The reduction depends on each site’s contribution to the total information, so unequal information across sites can produce unequal attenuation. This provides a working-model explanation for the downward bias of the sandwich variance with few clusters [16, 17]. The score-space corrections below account for this attenuation through site leverage; Section 2.7 states the conditions under which the correction is exact.

Remark 1 (Independent units in the VHA analysis). The VHA cohort contains 29,041 hospitalizations, but the site-level sandwich is constructed from only 18 VISN scores. Additional hospitalizations can increase information for estimation without increasing the number of independent units available for variance estimation. With $p = 12$ coefficients, the rank bound does not force singularity, but neither the large cohort nor $K > p$ ensures that the large-sample reference distribution is adequate.

2.5 Score-space correction and site leverage

Conventional small-sample corrections use observation-level residuals and cluster leverages [16–18]. In a federation, the coordinating center instead has access to $\widehat{S}_i$ and $\mathcal{B}_i(\widehat{\beta})$. We express the corrections through these summaries, allowing variance estimation to be completed in one additional exchange after convergence.

Define the information-standardized score and site leverage

$$z_i = \hat{\mathcal{B}}^{-1/2} \hat{\mathcal{S}}_i, \quad G_i = \hat{\mathcal{B}}^{-1/2} \mathcal{B}_i(\hat{\beta}) \hat{\mathcal{B}}^{-1/2}. \quad (6)$$

Each G_i is symmetric and positive semidefinite, with $\sum_i G_i = \mathbf{I}$ and $\sum_i \text{tr}(G_i) = p$. It measures site i 's contribution to the total information and serves as the site leverage. For linear regression with one observation per site and $V_{ij} = 1$, $\text{tr}(G_i)$ reduces to the usual hat value h_{ii} . If all sites contribute equal information, $G_i = \mathbf{I}/K$; otherwise, leverage varies with site size and covariate distribution. Web Appendix C establishes invariance to the choice of matrix square root.

We require $\mathbf{I} - G_i \succ 0$ for every site, or equivalently $\hat{\mathcal{B}} - \mathcal{B}_i(\hat{\beta}) \succ 0$. The condition fails only if one site alone identifies some direction of β . We report the smallest eigenvalue of $\mathbf{I} - G_i$ across sites as a diagnostic.

2.6 Variance corrections

Kauermann–Carroll (KC) and Mancl–DeRouen (MD). For an exponent $\alpha > 0$, let $A_i(\alpha) = (\mathbf{I} - G_i)^{-\alpha}$ and

$$\hat{V}(\alpha) = \hat{\mathcal{B}}^{-1/2} \left\{ \sum_{i=1}^K A_i(\alpha) z_i z_i^\top A_i(\alpha) \right\} \hat{\mathcal{B}}^{-1/2}. \quad (7)$$

The adjustment inflates each standardized score according to its site leverage. The choice $\alpha = 1/2$ is the score-space analogue of the KC correction [17]. The choice $\alpha = 1$ is the analogue of the MD correction [16]. We write $\hat{V}_{\text{KC}} = \hat{V}(1/2)$ and $\hat{V}_{\text{MD}} = \hat{V}(1)$. Kauermann and Carroll [17] chose the half-power so that the corrected variance is unbiased in the linear model with a correct working covariance. Mancl and DeRouen [16] derived the full power from a first-order bias approximation. Under the working model of Section 2.7, the full power overestimates the variance (Theorem 1). The adjustment acts on the standardized score. Applying $(\mathbf{I} - G_i)^{-\alpha}$ directly to the raw score $\hat{\mathcal{S}}_i$ gives a different estimator in general (Web Appendix B).

Fay–Graubard (FG). Fay and Graubard [18] rescale each cluster's score by a diagonal factor. In score space the factor for site i is

$$F_i = \text{diag} \left[\left\{ 1 - \min \left(b, (\mathcal{B}_i \hat{\mathcal{B}}^{-1})_{kk} \right) \right\}^{-1/2} \right]_{k=1}^p, \quad \hat{V}_{\text{FG}} = \hat{\mathcal{B}}^{-1} \left(\sum_{i=1}^K F_i \hat{\mathcal{S}}_i \hat{\mathcal{S}}_i^\top F_i \right) \hat{\mathcal{B}}^{-1}, \quad (8)$$

with the default bound $b = 0.75$. FG inflates each coordinate of the score separately using the diagonal of the site's information share. The bound limits inflation when information about a coefficient is concentrated in one site. Li and Redden [20] found FG preferable to

201 KC when cluster sizes vary widely.

202 All three estimators share a common form. Write the corrected raw score as $T_i \hat{\mathcal{S}}_i$, with
 203 $T_i = \hat{\mathcal{B}}^{1/2} A_i(\alpha) \hat{\mathcal{B}}^{-1/2}$ for KC and MD and $T_i = F_i$ for FG. Then each estimator equals

$$\hat{V}_T = \hat{\mathcal{B}}^{-1} \left(\sum_{i=1}^K T_i \hat{\mathcal{S}}_i \hat{\mathcal{S}}_i^\top T_i^\top \right) \hat{\mathcal{B}}^{-1}. \quad (9)$$

204 The uncorrected sandwich (4) is the case $T_i = \mathbf{I}$.

205 *Remark 2* (Relation to CR2). With working independence and an identity link, $\hat{V}_{\text{KC}}$ equals
 206 the bias-reduced linearization estimator CR2 [19, 22]. For logistic GEE the two differ,
 207 because the working weights are not constant. In the VHA application they differed by less
 208 than 0.5% (Section 4).

209 2.7 Exactness of the half-power

210 The half-power in the KC correction is exact under a working model for the scores. The
 211 model treats $\hat{\mathcal{B}}$ and the $\mathcal{B}_i$ as fixed matrices B and B_i .

212 **Theorem 1** (Affine working-score exactness). *Conditionally on the design, suppose $\mathcal{S}_i(\beta) =$
 213 $U_i - B_i(\beta - \beta_0)$, where the B_i are fixed positive semidefinite matrices, the U_i are independent,
 214 $E(U_i) = 0$, and $\text{Var}(U_i) = \tau B_i$ for some $\tau > 0$. Suppose $B = \sum_i B_i \succ 0$ and $B - B_i \succ 0$ for
 215 every i . Define G_i and z_i by (6) with $\hat{\mathcal{B}}$ replaced by B . Then*

$$E(z_i z_i^\top) = \tau G_i (\mathbf{I} - G_i), \quad E \left\{ \sum_{i=1}^K A_i(\tfrac{1}{2}) z_i z_i^\top A_i(\tfrac{1}{2}) \right\} = \tau \mathbf{I}.$$

216 Hence $\hat{V}_{\text{KC}}$ is unbiased for $\text{Var}(\hat{\beta}) = \tau B^{-1}$ at every finite K . Within the family $A_i(\alpha)$,
 217 $\alpha = 1/2$ is the only exponent that is unbiased for all admissible $\{B_i\}$. The choice $\alpha = 1$
 218 overestimates the variance.

219 The proof is given in Web Appendix B. Under the affine working model, the half-power
 220 adjustment removes the attenuation in the expected outer product of each standardized
 221 score. The condition $\text{Var}(U_i) = \tau B_i$ requires the working covariance to be correct up to a
 222 common scale, as in the original KC derivation [17]. The result establishes exactness and
 223 uniqueness of the exponent within this family under that condition. For logistic GEE or
 224 a misspecified working covariance, the correction is based on linearization; its finite-sample
 225 performance is evaluated in Section 3.

226 *Remark 3.* As $K \rightarrow \infty$ with no dominant site, $\max_i \lambda_{\max}(G_i) \rightarrow 0$ in probability. Then
 227 $\hat{V}_{\text{KC}}$, $\hat{V}_{\text{MD}}$, $\hat{V}_{\text{FG}}$, and $\hat{V}_{\text{S}}$ have the same first-order limit (Web Appendix D). Their effects are

therefore asymptotically negligible under these conditions.

2.8 Reference distribution

Small-sample inference also requires an appropriate reference distribution. The Wald statistic $(\hat{\beta}_j - \beta_j)/\hat{V}_{jj}^{1/2}$ is compared with a reference distribution, and the normal distribution is too light-tailed when K is small [20]. Common choices are the t distribution with $K - p$ or $K - 1$ degrees of freedom [16, 23]. A fixed count does not respond to unequal information across sites. We therefore use the Satterthwaite approximation of Bell and McCaffrey [19], computed in score space.

The variance estimator is constructed from a small number of site scores and can remain highly variable after bias correction [17]. The Bell–McCaffrey degrees of freedom account for this variability by matching the first two moments of the corrected variance under a working model. Equal information across sites gives $K - 1$ degrees of freedom; concentration of information in fewer sites reduces the effective degrees of freedom and increases the critical value.

Fix a coefficient j and let e_j be the j th unit vector. Define $a_i = T_i^\top \hat{\mathcal{B}}^{-1} e_j$. By (9), the corrected variance of $\hat{\beta}_j$ is

$$\hat{V}_{T,jj} = \sum_{i=1}^K (a_i^\top \hat{\mathcal{S}}_i)^2.$$

Under the working model of Theorem 1 with Gaussian U_i , this is a quadratic form in the U_i . Its first two moments are determined by the $K \times K$ matrix

$$C_{ik} = \mathbf{1}\{i = k\} a_i^\top \mathcal{B}_i a_i - (\mathcal{B}_i a_i)^\top \hat{\mathcal{B}}^{-1} (\mathcal{B}_k a_k), \quad i, k = 1, \dots, K. \quad (10)$$

Matching these moments to a scaled χ^2 distribution gives the degrees of freedom

$$\nu_j^{\text{BM}} = \frac{\{\text{tr}(C)\}^2}{\text{tr}(C^2)}. \quad (11)$$

The interval for β_j is $\hat{\beta}_j \pm t_{\nu_j^{\text{BM}}, 0.975} \hat{V}_{T,jj}^{1/2}$.

The degrees of freedom are computed from the fitted information matrices $\mathcal{B}_i$ and correction operators T_i , without additional outcome summaries. For nonlinear models, these matrices can depend on the outcomes through $\hat{\beta}$. When all sites carry equal information, $\nu_j^{\text{BM}} = K - 1$. When information concentrates in a few sites, ν_j^{BM} falls below $K - 1$, and it can differ across coefficients. This matches the design-based logic of Bell and McCaffrey [19] and Imbens and Kolesár [21]. With working independence and an identity link, ν_j^{BM} equals the CR2 Satterthwaite degrees of freedom of Pustejovsky and Tipton [22]. Web Appendix

F gives the derivation. For FG we also report the degrees of freedom proposed by Fay and Graubard [18] as a sensitivity analysis. Our recommended procedure is $\hat{V}_{KC}$ with a $t_{\nu_j^{BM}}$ reference distribution. Section 3 gives the evidence for this choice.

Remark 4 (Coefficient-specific information across VISNs). As discussed later in the VHA analysis, the Bell–McCaffrey degrees of freedom are 7.4 for non-clinician consultation and 14.6 for psychiatry discharging service (Table 3). Although both coefficients use the same 18 VISNs, information for the consultation contrast is more concentrated across sites. A fixed t_{17} reference would assign the same critical value to both, whereas the coefficient-specific reference assigns a larger critical value to the consultation contrast. The alternative fixed count $K - p = 6$ would give wider intervals than the Bell–McCaffrey reference for both coefficients, holding the variance estimates fixed.

2.9 Federated algorithm with a coordinating center

Algorithm 1 combines iterative estimation with the score-space correction and reference distribution. Once fitting converges, each site evaluates its score and information matrix at $\hat{\beta}$ and sends these summaries to the center. The center then computes the corrected variance and coefficient-specific degrees of freedom. This final stage requires no further parameter updates and no additional type of patient-level summary.

Algorithm 1 Fed-GEE with a coordinating center and one-shot small-sample correction

Require: Local datasets at K sites; common initial value $\beta^{(0)}$; working covariance specification; tolerance $\epsilon = 10^{-8}$; maximum iterations $M = 50$; correction KC, MD, or FG.

Ensure: Estimate $\hat{\beta}$, corrected variance $\hat{V}_T$, and coefficient-specific degrees of freedom ν_j^{BM} .

- 1: **Iterative federated fitting**
 - 2: **for** $r = 0, \dots, M - 1$ **do**
 - 3: Center broadcasts $\beta^{(r)}$ to all sites.
 - 4: Each site i computes and sends $\mathcal{S}_i\{\beta^{(r)}\}$ and $\mathcal{B}_i\{\beta^{(r)}\}$ using (2), in parallel.
 - 5: Center sums the scores and information matrices and updates $\beta^{(r+1)}$ using (3).
 - 6: **if** $\|\beta^{(r+1)} - \beta^{(r)}\| < \epsilon$ **then**
 - 7: Set $\hat{\beta} = \beta^{(r+1)}$ and exit the loop.
 - 8: **end if**
 - 9: **end for**
 - 10: If convergence was not reached, report nonconvergence and stop.
 - 11: **One-shot post-convergence correction**
 - 12: Center broadcasts $\hat{\beta}$; each site recomputes and sends $\hat{\mathcal{S}}_i$ and $\mathcal{B}_i(\hat{\beta})$ once.
 - 13: Center forms $\hat{\mathcal{B}}$, checks the leverage conditions in Section 2.5, and constructs the chosen operators T_i from Section 2.6.
 - 14: Center computes $\hat{V}_T$ using (9) and ν_j^{BM} using (10)–(11).
 - 15: **return** $\hat{\beta}$, $\hat{V}_T$, and $\{\nu_j^{BM}\}_{j=1}^p$; form intervals and tests using the corresponding t reference distributions.
-

2.10 Decentralized estimation and inference

When a coordinating center is unavailable, sites can exchange summaries with their neighbors and approximate the required network totals through consensus. Related work by Gu and Chen [24] develops inference for decentralized M-estimation using stochastic-gradient updates and Polyak–Ruppert averaging. Here, we use consensus on GEE scores and information matrices to approximate the Fisher-scoring update and extend the score-space correction to this architecture.

Let $W = (w_{ih})$ be a nonnegative, symmetric, doubly stochastic mixing matrix on a connected network, with $w_{ih} = 0$ for non-neighboring sites. Write $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and assume $\rho = \|W - J_K\|_2 < 1$. For a collection of local vectors or matrices $Q_1, \dots, Q_K$, define

$$\mathcal{A}_{L,i}(Q) = \sum_{h=1}^K [W^L]_{ih} Q_h. \quad (12)$$

This quantity is computed through L rounds of neighbor exchange, with $Q_i^{[\ell+1]} = \sum_h w_{ih} Q_h^{[\ell]}$. As L increases, $\mathcal{A}_{L,i}(Q)$ approaches the network average, and $K\mathcal{A}_{L,i}(Q)$ approaches the sum [25]. For symmetric W , $\|W^L - J_K\|_2 = \rho^L$; the corresponding maximum row-sum error is bounded by $\sqrt{K}\rho^L$. Communication requirements therefore depend on both network connectivity and the number of sites.

Each site maintains a parameter vector $\beta_i^{(r)}$. At each iteration, sites first average their parameter states through consensus, evaluate their local scores and information matrices, and then average these summaries. Writing the resulting quantities as $\tilde{\beta}_i^{(r)}$, $\bar{\mathcal{S}}_i^{(r)}$, and $\bar{\mathcal{B}}_i^{(r)}$, the update is

$$\beta_i^{(r+1)} = \tilde{\beta}_i^{(r)} + \eta_r \{\bar{\mathcal{B}}_i^{(r)}\}^{-1} \bar{\mathcal{S}}_i^{(r)}, \quad (13)$$

where $\eta_r > 0$ is a step size. The factor K cancels between the averaged score and information matrix. After terminal parameter consensus, denote the local estimates by $\hat{\beta}_i$ and their network average by $\hat{\beta}_{\text{dec}}$.

Web Appendix H establishes consistency when parameter-consensus error, score-consensus error, and the terminal estimating-equation residual vanish. If their combined error is $o(K^{-1/2})$, then $\hat{\beta}_{\text{dec}}$ has the same first-order distribution as the estimator obtained with a coordinating center. These conditions require sufficient communication and numerical convergence. With $W = J_K$, a single round gives exact averages and, under common initialization and unit step sizes, reproduces the coordinated update.

The small-sample correction is performed after fitting. Each site evaluates $\mathcal{S}_i(\hat{\beta}_i)$ and $\mathcal{B}_i(\hat{\beta}_i)$, then uses consensus to obtain an approximation $\hat{\mathcal{B}}^{(i)}$ to the total information. From this matrix and its local information, site i constructs its correction operator T_i and corrected

meat contribution $T_i \hat{\mathcal{S}}_i \hat{\mathcal{S}}_i^\top T_i^\top$. A further consensus stage aggregates these contributions. Under the smoothness and eigenvalue conditions in Web Appendix H, the corrected variance approaches its coordinated counterpart as parameter and summary consensus errors vanish.

The Bell–McCaffrey degrees of freedom can also be computed from additive summaries. For each coefficient, let $a_i = T_i^\top \hat{\mathcal{B}}^{-1} e_j$, $d_i = a_i^\top \mathcal{B}_i a_i$, and $u_i = \mathcal{B}_i a_i$. The traces in (11) depend only on $\sum_i d_i$, $\sum_i d_i^2$, $\sum_i u_i u_i^\top$, and $\sum_i d_i u_i u_i^\top$, together with $\hat{\mathcal{B}}^{-1}$. Web Appendix H gives the identities. Consensus approximations to these sums therefore provide coefficient-specific reference distributions without collecting patient-level records.

Algorithm 2 summarizes this extension. The correction is one-shot in the sense that it is a single post-fitting stage; it requires sequential consensus operations, each of which may involve several neighbor exchanges.

Algorithm 2 Decentralized Fed-GEE with a post-convergence score-space correction

Require: Local datasets; mixing matrix W ; initial states $\beta_i^{(0)}$; step sizes η_r ; consensus budgets $L_\beta, L_S, L_B, L_M, L_\nu$; stopping tolerances and iteration limit; correction KC, MD, or FG.

Ensure: Local approximations to the coordinated estimate, corrected variance, and Bell–McCaffrey degrees of freedom.

- 1: **Iterative decentralized fitting**
 - 2: **for** $r = 0, 1, \dots$ up to the iteration limit **do**
 - 3: Each site sets $\tilde{\beta}_i^{(r)} = \mathcal{A}_{L_\beta, i}(\beta^{(r)})$.
 - 4: Each site computes $\mathcal{S}_i(\tilde{\beta}_i^{(r)})$ and $\mathcal{B}_i(\tilde{\beta}_i^{(r)})$ using (2).
 - 5: Apply $\mathcal{A}_{L_S, i}$ to scores and $\mathcal{A}_{L_B, i}$ to information matrices to obtain $\bar{\mathcal{S}}_i^{(r)}$ and $\bar{\mathcal{B}}_i^{(r)}$.
 - 6: Update each $\beta_i^{(r+1)}$ using (13).
 - 7: At a candidate stopping point, perform terminal parameter consensus and recompute score and information averages.
 - 8: Stop when all sites satisfy the terminal residual and agreement tolerances of Web Appendix H, using neighbor communication to confirm.
 - 9: **end for**
 - 10: If convergence was not reached, report nonconvergence and stop.
 - 11: **Post-convergence correction**
 - 12: Each site computes $\hat{\mathcal{S}}_i = \mathcal{S}_i(\hat{\beta}_i)$, $\mathcal{B}_i = \mathcal{B}_i(\hat{\beta}_i)$, and $\hat{\mathcal{B}}^{(i)} = K \mathcal{A}_{L_B, i}(\mathcal{B})$.
 - 13: Check leverage conditions and construct T_i using $\hat{\mathcal{B}}^{(i)}$ and $\mathcal{B}_i$ as in Section 2.6.
 - 14: Form $M_i = T_i \hat{\mathcal{S}}_i \hat{\mathcal{S}}_i^\top T_i^\top$ and set $\hat{V}_T^{(i)} = (\hat{\mathcal{B}}^{(i)})^{-1} \{K \mathcal{A}_{L_M, i}(M)\} (\hat{\mathcal{B}}^{(i)})^{-1}$.
 - 15: For each coefficient j , form d_i, u_i using $a_i = T_i^\top (\hat{\mathcal{B}}^{(i)})^{-1} e_j$; apply $K \mathcal{A}_{L_\nu, i}$ to $d_i, d_i^2, u_i u_i^\top, d_i u_i u_i^\top$.
 - 16: Compute local trace approximations and $\nu_j^{\text{BM}, (i)}$ using Web Appendix H; increase consensus budgets if the required positivity conditions fail.
 - 17: **return** $\hat{\beta}_i, \hat{V}_T^{(i)}$, and $\{\nu_j^{\text{BM}, (i)}\}_{j=1}^p$ at each site.
-

3 Simulation study

The simulations compare variance corrections and reference distributions when sites are few. We varied the number of sites, the model dimension, the outcome prevalence, and the balance of site sizes. Unequal site sizes are common in health systems. They lower the Bell–McCaffrey degrees of freedom but leave fixed counts such as $K - 1$ and $K - p$ unchanged. We also compared Fed-GEE with pooled and meta-analytic GEE.

3.1 Design

We generated clustered binary outcomes with a marginal logistic mean. Visit t of patient j at site i had a covariate $x_{ijt} \sim N(0, 1)$ and marginal probability $\text{logit}^{-1}(\beta_0 + \beta_x x_{ijt})$, with $\beta_x = 1$. Outcomes were produced by thresholding a latent Gaussian variable,

$$Z_{ijt} = \sqrt{\rho_H} U_i + \sqrt{\rho_P - \rho_H} W_{ij} + \sqrt{1 - \rho_P} \varepsilon_{ijt},$$

with independent standard normal U_i , W_{ij} , and ε_{ijt} . This construction fixes the marginal mean exactly and induces correlation within patients ($\rho_P = 0.30$) and within sites ($\rho_H = 0.05$) on the latent scale. Each patient had two to six visits.

The design varied four factors. The number of sites was $K \in \{10, 18, 30, 50\}$, which includes the $K = 18$ of the application. The model dimension was $p \in \{2, 5, 10, 20\}$: an intercept, x , and up to 18 nuisance covariates. When present, the first nuisance covariate was a binary site-level variable. The rest were visit-level continuous and binary variables. We kept scenarios with $K - p \geq 5$, which leaves 13 combinations of K and p . The intercept $\beta_0 \in \{-3, -1, 0\}$ gave average outcome prevalences of 7%, 30%, and 50%. The VHA prevalence of 30.8% lies at the middle level. Site sizes had one of three distributions, all with a mean of about 20 patients. In the equal-size arm, sizes were uniform on 10 to 30 patients, a coefficient of variation (CV) of 0.30. In two unequal arms, sizes were gamma distributed with CV 0.78 and 1.00 and a minimum of five patients. Holding the mean fixed separates the effect of imbalance from the effect of total sample size. The full design had $13 \times 3 \times 3 = 117$ scenarios, each with 1,000 replicates.

Every replicate was fitted by federated GEE with working exchangeability. We computed the uncorrected site sandwich (4) and the KC, MD, and FG corrections of Section 2.6. Each variance was paired with four reference distributions: normal, t_{K-1} , t_{K-p} , and $t_{\nu\text{BM}}$. FG was also paired with its own degrees of freedom [18]. We also fitted two comparators. Pooled GEE used all data in one place, with the site-clustered sandwich. Meta-analytic GEE fitted a separate GEE at each site and combined the site estimates of β_x by inverse-variance

weighting. Site-level covariates cannot be estimated within a site and were omitted from the local fits. Sites whose local fit failed or showed separation were excluded.

The target was β_x . We report the coverage of nominal 95% intervals and the ratio of the mean estimated standard error to the empirical standard deviation of $\hat{\beta}_x$ (SE/SD), for which one is ideal. With 1,000 replicates the Monte Carlo standard error of a coverage near 0.95 is 0.007. Table 1 shows the 12 scenarios closest to the VHA application: prevalence 30% and the largest p at each K , which gives the smallest $K - p$ to guarantee non-singularity of the meat matrix. Web Tables S3 and S4 report all 117 scenarios. We summarize results as ranges over scenarios, without averaging.

3.2 Results

The uncorrected sandwich with a normal reference undercovered. Across the 117 scenarios its coverage ranged from 0.863 to 0.952, and 79 scenarios fell below 0.93. The problem was worst with few and unequal sites. At $K = 10$ with CV 0.78 or 1.00, coverage was 0.876 (Table 1). This agrees with earlier work on few clusters [16, 20].

The reference distribution did much of the repair. Keeping the uncorrected variance but using $t_{\nu\text{BM}}$ raised coverage to 0.920 to 0.962. The remaining gap came from the biased variance, with SE/SD as low as 0.81. Adding the KC correction closed it. KC with $t_{\nu\text{BM}}$ covered at 0.929 to 0.971, and only one scenario fell below 0.93. It was within 0.01 of 0.95 in 94 of the 117 scenarios. The corresponding counts were 74 for KC with t_{K-p} , 67 for KC with t_{K-1} , 72 for MD with $t_{\nu\text{BM}}$, and 16 for the uncorrected sandwich with a normal reference. The range was similar at each prevalence: 0.929 to 0.965, 0.935 to 0.971, and 0.938 to 0.969 for prevalences 7%, 30%, and 50%. In the scenario closest to the application ($K = 18$, $p = 10$), it covered at 0.949 to 0.958 across the three site-size distributions, compared with 0.902 to 0.933 for the uncorrected sandwich.

The choice of degrees of freedom mattered consequentially when sites were unequal. With equal sites, ν^{BM} was close to $K - 1$. With unequal sites it fell well below $K - 1$. At $K = 18$ it was 15.3, 10.9, and 9.0 for CV of 0.30, 0.78, and 1.00. A fixed t_{K-1} reference ignores this and undercovered with unequal sites, at 0.934 for $K = 10$ with CV of 0.78 and 0.935 for $K = 18$ with CV 1.00. The count $K - p$ also ignores imbalance, and it depends on p . When $K - p$ was small it overcovered, up to 0.973 in Table 1.

The other corrections behaved as the theory suggests. MD overcorrected, with SE/SD from 0.95 to 1.09, as Theorem 1 predicts for the full power. With $t_{\nu\text{BM}}$ it overcovered, up to 0.983. FG was almost identical to KC. The two coverages differed by at most 0.005 in any scenario, and the FG degrees of freedom were close to ν^{BM} (Web Table S3). In our design

the largest site leverage averaged at most 0.44, so the FG bound rarely applied.

Federated and pooled GEE gave the same estimates and standard errors in every replicate. The largest differences were 3×10^{-15} in $\hat{\beta}_x$ and a relative 10^{-8} in the standard error.

Table 2 compares the recommended Fed-GEE procedure with meta-analytic GEE. Fed-GEE covered at 0.929 to 0.971 across all scenarios, and its bias stayed below 0.3 empirical standard deviations. Meta-analytic GEE covered at 0.010 to 0.912. Its standard errors were too small, and its bias grew with K and p , up to 3.68 empirical standard deviations. Each local fit uses about 20 patients. A logistic fit on so few patients is biased away from zero [26]. In our simulations this bias was larger when the local model had more covariates. Averaging many such fits keeps the bias while the standard error shrinks. Sites excluded for failed or separated local fits add to the problem. The largest values of p are therefore a hard setting for meta-analytic GEE. Fed-GEE solves one pooled estimating equation and does not face this issue.

These results suggest the following recommendation: use the KC correction with a $t_{\nu\text{BM}}$ reference distribution. With equal sites, this is close to KC with t_{K-1} . With unequal sites, the Bell–McCaffrey degrees of freedom guard against the undercoverage of a fixed count. The recommendation held for K from 10 to 50, p up to 20, and outcome prevalence from 7% to 50%.

3.3 Decentralized implementation

A separate simulation evaluated the effect of network communication on decentralized estimation and inference (Web Appendix I). It used 5 to 100 sites, four network structures, and 50 rounds per consensus stage. Intervals used the uncorrected site sandwich with a t_{K-p} reference. These results therefore assess consensus approximation rather than the post-convergence small-sample correction in Algorithm 2.

On complete and regional networks, decentralized coverage closely matched the corresponding coordinated analysis. On a single-hub network with $K = 100$, where $\rho^{50} = 0.61$, coverage ranged from 0.06 to 0.58. Although the slope estimate remained within 0.1 of its true value, its error was substantial relative to the estimated standard error. The results show that a fixed communication budget can be adequate for one network and inadequate for another. Web Appendix I reports the full design and results.

4 VHA application

In the cohort described in the Introduction, MAUD initiation occurred in 8,932 hospitalizations (30.8%). Clinician consultation occurred in 834 hospitalizations (2.9%) and non-clinician consultation in 4,354 (15.0%) [9]. We fitted model (1), emulating federation across VISNs using centrally available records and checking the results against a pooled analysis.

The number of Veterans per VISN had a CV of 0.35, close to the equal-size arm of the simulation (CV 0.30). The largest VISN leverage, the largest diagonal element of $\mathcal{B}_i \hat{\mathcal{B}}^{-1}$, was 0.27. The smallest eigenvalue of $\mathbf{I} - G_i$ across VISNs was 0.725, so every correction was well defined and none required regularization. The FG bound was never reached. The closest simulation scenario has $K = 18$, $p = 10$, prevalence 0.30, and CV 0.30 (Table 1). There, the uncorrected sandwich with a normal reference covered at 0.933 and the recommended procedure at 0.958. With more imbalance (CV 0.78) the corresponding values were 0.921 and 0.957.

Table 3 and Figure 1 report the results. Both consultation types were associated with higher MAUD initiation. Clinician consultation had an odds ratio of 1.47 (95% CI, 1.14 to 1.90). Non-clinician consultation had an odds ratio of 1.88 (95% CI, 1.24 to 2.84). The two intervals overlap, and we did not test whether the two odds ratios differ. The results therefore support both consultation types as options for expanding consultation capacity. They do not show that one type is more effective.

The recommended procedure widened every interval relative to pooled GEE with the uncorrected sandwich. On the log scale the widening ranged from 13% to 42%. Most of it came from the reference distribution. The KC correction raised the standard errors by 4% to 12%, and the $t_{\nu\text{BM}}$ critical values were 9% to 28% larger than 1.96. A small correction is expected here. The VISNs are nearly equal in size, so no VISN holds a large share of the information. No association changed statistical significance. The associations with psychiatry discharging service, opioid use disorder, Care Assessment Need score, and frailty remained clear.

The Bell–McCaffrey degrees of freedom ranged from 5.5 to 14.7 across coefficients (Table 3). The lowest values were for discharge against medical advice and non-clinician consultation, whose intervals widened by 42% and 29%, respectively. Remark 4 relates this variation to the choice of reference distribution.

Two checks support the calculation. First, CR2 with Satterthwaite degrees of freedom, computed on the pooled data with the `clubSandwich` package [22], gave standard errors within 0.5% of the federated KC values and the same degrees of freedom to three decimal places. Second, FG with $t_{\nu\text{BM}}$ reproduced the KC interval limits to within 0.01. The un-

corrected sandwich with $t_{\nu\text{BM}}$, FG with its own degrees of freedom, and CR2 gave the same conclusions for every coefficient (Web Table S5).

Meta-analytic GEE gave different point estimates. The odds ratio for female sex was 1.04 (95% CI, 0.88 to 1.24) compared with 1.22 under GEE. The odds ratio for intensive care unit stay was 0.84 (95% CI, 0.72 to 0.99) compared with 0.88. These differences change the reported significance of both contrasts. Meta-analytic GEE does not target the pooled estimand. It weights VISN-specific estimates, and VISNs whose local fit fails may be excluded. Fed-GEE reproduces the pooled estimate exactly and keeps every VISN in the analysis.

5 Discussion

In a federated analysis of 29,041 hospitalizations in 18 VISNs, both clinician and non-clinician addiction consultation were associated with higher MAUD initiation, after accounting for regional clustering and the small number of VISNs. The odds ratios were 1.47 (95% CI, 1.14 to 1.90) and 1.88 (95% CI, 1.24 to 2.84).

The methodological contribution addresses what remains after federated GEE has reproduced the pooled estimate. With few sites, the sandwich variance is too small and normal critical values are too narrow. The standard corrections cannot be computed by a coordinating center in their usual form. We showed that they can be written in terms of the site scores and information matrices alone. The same holds for the Bell–McCaffrey degrees of freedom. The calculation adds one post-convergence exchange of summaries that the sites already send.

The simulations support one procedure: the KC correction with a $t_{\nu\text{BM}}$ reference distribution. Two findings explain why. First, the reference distribution does much of the work. The uncorrected sandwich with $t_{\nu\text{BM}}$ already came close to nominal coverage, and KC closed the remaining gap. This agrees with Kauermann and Carroll [17], who showed that the variability of the sandwich lowers coverage even when its bias is removed. Second, fixed degrees of freedom fail when sites are unequal. The Bell–McCaffrey value equals $K - 1$ for equal sites and falls as information concentrates, while $K - 1$ and $K - p$ do not change. Li and Redden [20] found FG preferable to KC when cluster sizes vary widely. In our design, FG and KC were nearly identical, including at a site-size CV of 1.00. One possible reason is that both corrections are here defined at the site level from the same information shares, but we did not test this explanation.

The application shows the practical value of the procedure. The correction widened every interval, by 13% to 42% on the log scale, but changed no conclusion. The Bell–McCaffrey degrees of freedom identified the contrasts whose information was concentrated in fewer

VISNs. These were non-clinician consultation and discharge against medical advice. Their intervals widened the most. A fixed reference distribution would have treated all contrasts alike.

For VHA policy, the results support non-clinician consultation as a way to expand consultation capacity. Non-clinician consultation was more common than clinician consultation in the cohort, occurring in 15.0% and 2.9% of hospitalizations [9]. Its estimated association with MAUD initiation was larger than that of clinician consultation, although we did not test the difference. These findings agree with the source cohort analysis, which used hospital fixed effects and patient-clustered standard errors [9]. They are also consistent with evidence on addiction consultation services for opioid use disorder [6, 7] and with the more limited evidence for AUD [8]. They are associations. Veterans who received a consultation may differ from those who did not in ways the model does not capture.

Fed-GEE targets marginal, population-averaged associations. A federated mixed model is preferable when the scientific question concerns conditional, site-specific effects [14]. One-shot distributed estimators reduce communication but approximate the pooled estimate [27]. Fed-GEE reproduces it exactly, at the cost of several rounds of communication. Web Appendix G gives a one-step version for settings with a good common starting value. When no coordinating center is available, the decentralized version of Section 2.10 applies. Its main practical demand is enough consensus rounds for the network at hand.

The correction also changes what must be shared. Estimation needs only the sums $\sum_i \mathcal{S}_i$ and $\sum_i \mathcal{B}_i$, which is compatible with secure aggregation. The correction needs each site’s score and information matrix separately. Any added privacy protection must therefore cover site-level summaries, not only their totals. This affects both the communication protocol and the trade-off between privacy and accuracy [15].

This study has limitations. First, the VHA analysis is observational, and the odds ratios are adjusted associations. Second, the records were available centrally, so federation was emulated. This allowed us to verify the federated results against a pooled analysis. Third, the exactness result holds only under an affine working model in which the working covariance is correct up to scale. For logistic GEE the correction is a linearization-based adjustment, and its performance rests on simulation. Fourth, the simulations used working independence, binary outcomes, and one data-generating mechanism with a fixed site-level correlation. Other outcomes, working correlations, and forms of heterogeneity need study. Fifth, we did not evaluate other small-sample approaches, such as the correction of Morel et al. [28] or the wild cluster bootstrap [29]. Finally, federation keeps patient-level records in place but does not by itself provide differential privacy or cryptographic security [12, 15].

In the VHA analysis, both clinician and non-clinician addiction consultation were asso-

ciated with higher MAUD initiation after accounting for regional clustering and the small number of VISNs. These findings support considering a broader consultation workforce while retaining access to clinicians who can prescribe treatment. Fed-GEE provides a way to address such staffing questions across health systems using site summaries: estimation proceeds through iterative communication, followed by a single post-convergence stage for variance correction and reference-distribution calculation. Although motivated by MAUD initiation, the framework applies more broadly to studies of correlated health outcomes where the target is a population-averaged association and patient-level records remain under local control.

Data availability statement

The patient-level VHA data contain sensitive health information and cannot be made publicly available. Access is subject to VHA data-governance and research approval procedures. The R package `FedGEE` implements federated GEE with the score-space corrections and Bell–McCaffrey degrees of freedom described in this paper. It is available at https://github.com/soumikp/2026_fedGee and can be installed with `remotes::install_github("soumikp/2026_fedGee")`. The same repository contains the code to reproduce the simulation study.

Ethics declaration

The institutional review board of the VA Pittsburgh Healthcare System approved the study and granted a waiver of informed consent [9].

Author contributions

Feier Chang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing. Soumik Purkayastha: Conceptualization, Methodology, Resources, Data curation, Supervision, Project administration, Writing – review and editing.

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Disclaimer

The content is solely the responsibility of the authors and does not necessarily represent the official views of the U.S. Department of Veterans Affairs, the Veterans Health Administration, or the United States Government.

Conflict of interest

The authors declare no conflicts of interest.

Use of generative artificial intelligence

The authors used Claude (Anthropic), through the Claude Code interface, to assist with manuscript drafting, auditing of the analysis and simulation code, and formatting of the FedGEE R package. All output was reviewed and revised by the authors, who take full responsibility for the accuracy, originality, and integrity of the manuscript and the software.

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657 **Tables**

Table 1: Coverage of nominal 95% confidence intervals for β_x in the simulation scenarios closest to the VHA application: outcome prevalence 0.30 and the largest p at each number of sites K . Each row is one scenario with 1,000 replicates. Web Table S3 reports all 117 scenarios.

K	p	CV	Uncorrected		KC			MD	FG	ν^{BM}
			z	$t_{\nu^{\text{BM}}}$	t_{K-1}	t_{K-p}	$t_{\nu^{\text{BM}}}$	$t_{\nu^{\text{BM}}}$	$t_{\nu^{\text{BM}}}$	
10	5	0.30	0.909	0.939	0.950	0.973	0.954	0.966	0.955	8.1
		0.78	0.876	0.933	0.934	0.957	0.951	0.962	0.947	5.8
		1.00	0.876	0.947	0.947	0.965	0.967	0.978	0.967	5.0
18	10	0.30	0.933	0.952	0.956	0.970	0.958	0.965	0.958	15.3
		0.78	0.921	0.950	0.952	0.966	0.957	0.967	0.956	10.9
		1.00	0.902	0.933	0.935	0.952	0.949	0.963	0.946	9.0
30	20	0.30	0.919	0.938	0.942	0.965	0.944	0.949	0.944	26.1
		0.78	0.935	0.956	0.954	0.973	0.964	0.970	0.961	18.2
		1.00	0.927	0.953	0.952	0.964	0.960	0.964	0.960	15.0
50	20	0.30	0.940	0.944	0.946	0.950	0.946	0.950	0.946	44.0
		0.78	0.932	0.943	0.944	0.946	0.949	0.949	0.947	30.5
		1.00	0.934	0.945	0.944	0.946	0.949	0.952	0.949	25.1

CV, coefficient of variation of site size. Uncorrected, site sandwich (4); KC, MD, FG, score-space corrections of Section 2.6. Column headings give the reference distribution. ν^{BM} , median Bell–McCaffrey degrees of freedom for KC. Monte Carlo standard error of each entry is about 0.007.

Table 2: Fed-GEE with the recommended procedure (KC with $t_{\nu_{\text{BM}}}$) and meta-analytic GEE in the scenarios of Table 1. Each row is one scenario. Web Table S4 reports all 117 scenarios.

K	p	CV	Fed-GEE			Meta-analytic GEE		
			Coverage	Bias/SD	SE/SD	Coverage	Bias/SD	SE/SD
10	5	0.30	0.954	0.16	0.988	0.892	0.36	0.858
		0.78	0.951	0.16	0.928	0.870	0.33	0.764
		1.00	0.967	0.21	0.956	0.867	0.36	0.752
18	10	0.30	0.958	0.19	1.012	0.685	1.22	0.844
		0.78	0.957	0.23	1.031	0.757	1.03	0.813
		1.00	0.949	0.19	0.939	0.757	0.86	0.756
30	20	0.30	0.944	0.13	0.978	0.135	2.49	0.741
		0.78	0.964	0.20	0.992	0.297	2.02	0.778
		1.00	0.960	0.23	0.977	0.385	1.81	0.785
50	20	0.30	0.946	0.18	0.984	0.028	3.15	0.724
		0.78	0.949	0.19	0.996	0.110	2.72	0.815
		1.00	0.949	0.15	0.947	0.190	2.32	0.786

CV, coefficient of variation of site size. Bias/SD, bias of $\hat{\beta}_x$ divided by its empirical standard deviation. SE/SD, mean estimated standard error divided by the empirical standard deviation. Meta-analytic GEE uses normal-reference intervals. Pooled GEE is not shown; it reproduced the uncorrected Fed-GEE estimates and standard errors in every replicate.

Table 3: Adjusted odds ratios (OR) and 95% confidence intervals for MAUD initiation in 29,041 hospitalizations of 18,900 Veterans in 18 VISNs.

		Pooled GEE	Fed-GEE (recommended)		Meta-analytic GEE
Covariate	OR	95% CI	95% CI	ν^{BM}	OR (95% CI)
<i>Demographics</i>					
Female	1.22	1.05, 1.41	1.03, 1.44	14.5	1.04 (0.88, 1.24)
Age ≥ 65 years	0.88	0.79, 0.98	0.78, 0.99	13.9	0.86 (0.76, 0.97)
<i>Alcohol severity</i>					
AUDIT-C no or low use	0.89	0.84, 0.93	0.83, 0.94	14.2	0.90 (0.81, 0.99)
Opioid use disorder	0.60	0.54, 0.68	0.53, 0.69	12.4	0.64 (0.57, 0.73)
<i>Medical complexity</i>					
CAN score ≥ 60 th percentile	0.72	0.65, 0.78	0.64, 0.79	14.7	0.73 (0.66, 0.80)
Frailty, mild or worse	0.69	0.61, 0.78	0.60, 0.79	12.7	0.65 (0.55, 0.77)
Any ICU stay	0.88	0.76, 1.02	0.74, 1.05	14.5	0.84 (0.72, 0.99)
<i>Hospital process</i>					
Psychiatry discharging service	1.73	1.43, 2.09	1.39, 2.15	14.6	1.83 (1.68, 1.99)
Addiction consult					
Clinician	1.47	1.19, 1.82	1.14, 1.90	10.4	1.66 (1.45, 1.90)
Non-clinician	1.88	1.36, 2.59	1.24, 2.84	7.4	1.73 (1.22, 2.46)
Discharge against advice	1.13	0.66, 1.92	0.53, 2.41	5.5	1.37 (0.83, 2.27)

Pooled GEE and Fed-GEE give the same OR. Pooled GEE uses the uncorrected VISN-clustered sandwich with a normal reference. Fed-GEE uses the score-space KC correction with a t reference on Bell–McCaffrey degrees of freedom ν^{BM} . Meta-analytic GEE fits a GEE in each VISN and pools the estimates by inverse-variance weighting. CAN, Care Assessment Need; ICU, intensive care unit; AUDIT-C, Alcohol Use Disorders Identification Test–Consumption. Reference categories: male; age 18–64 years; AUDIT-C unhealthy use or higher; no opioid use disorder; CAN score below the 60th percentile; nonfrail or prefrail; no ICU stay; medical or surgical discharging service; no addiction consult; routine discharge.

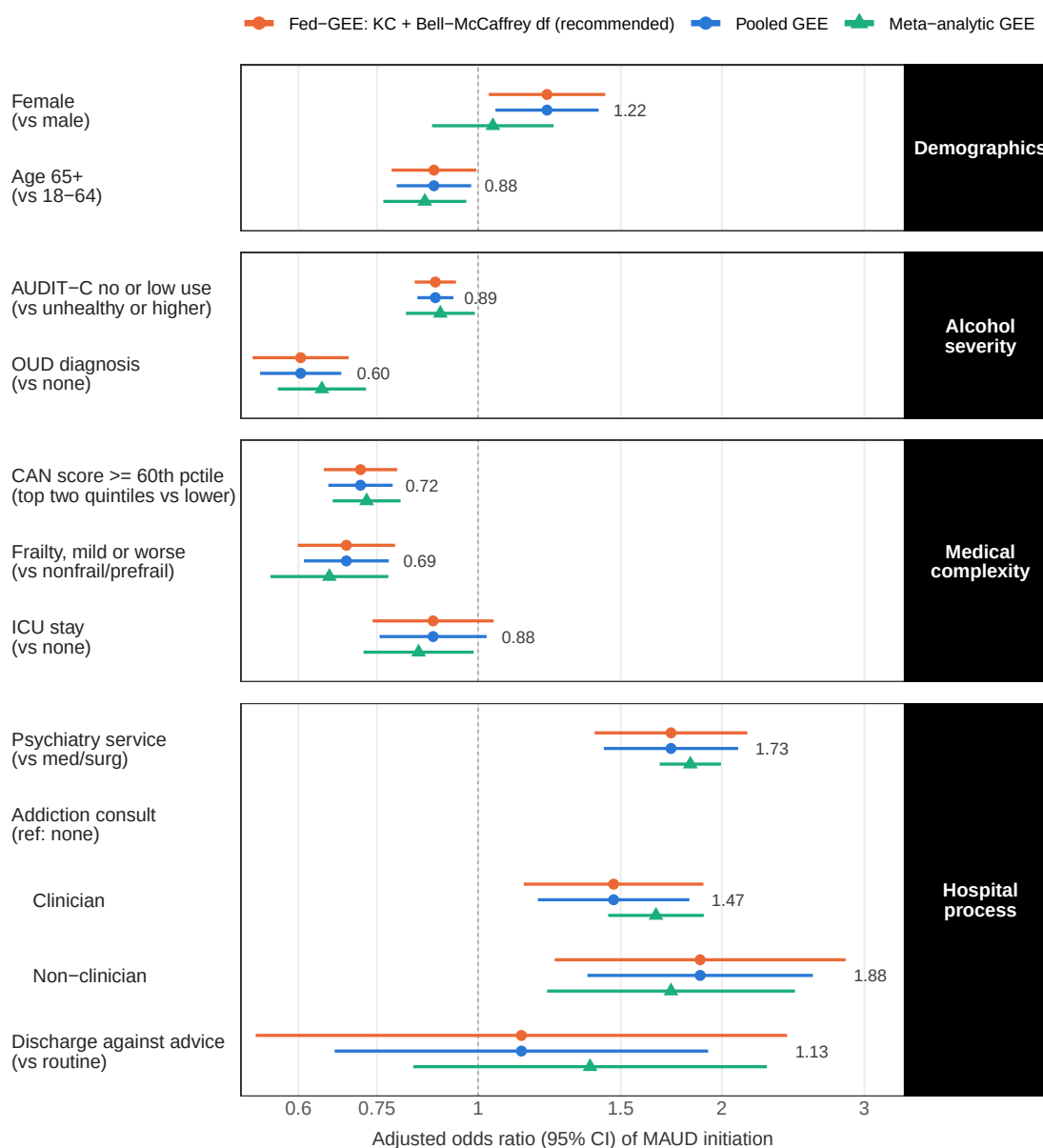
658 **Figure**

Figure 1: Adjusted odds ratios and 95% confidence intervals for MAUD initiation across 29,041 hospitalizations of 18,900 Veterans in 18 VISNs, the units of federation. Fed-GEE uses the score-space Kauermann–Carroll correction with a t reference distribution on Bell–McCaffrey degrees of freedom, computed from the VISN summaries alone. Pooled GEE uses the uncorrected VISN-clustered sandwich with a normal reference. Fed-GEE and pooled GEE give identical odds ratios and differ only in their intervals. Meta-analytic GEE fits a GEE in each VISN and pools the estimates by inverse-variance weighting. Numeric labels give the Fed-GEE odds ratio.